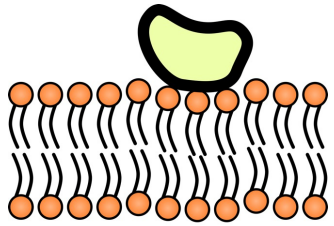
# C9555 A Survival Guide to Simulations

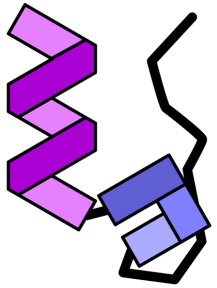
Sampling in Molecular Dynamics

Ivo Kabelka

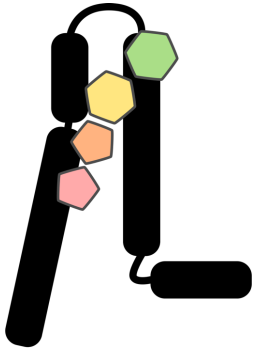
# Examples of studied processes



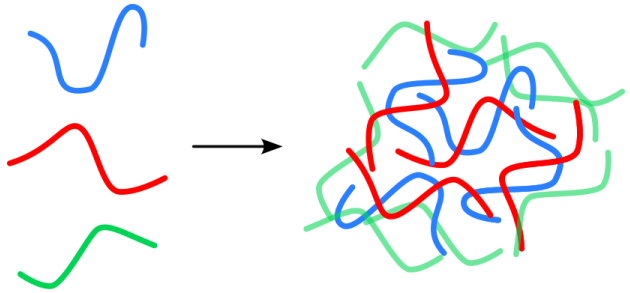
Membrane adsorption



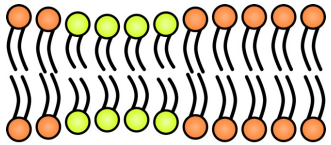
Conformational ensemble



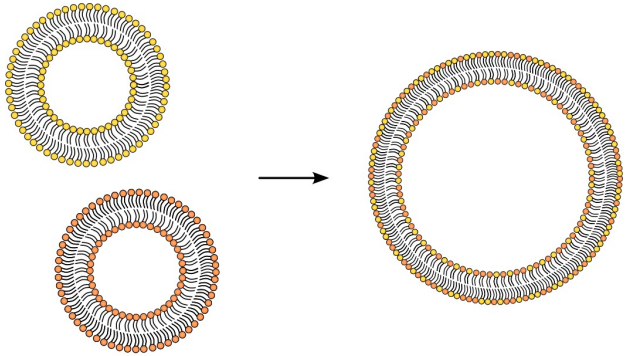
Ligand binding



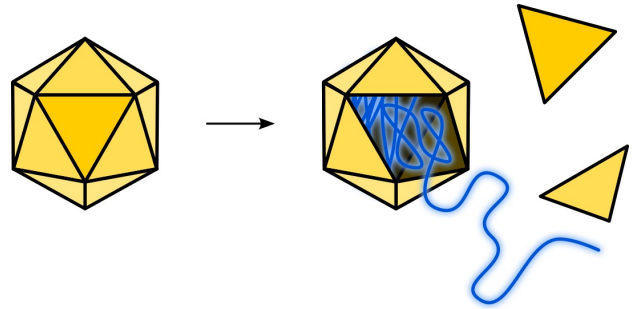
Aggregation



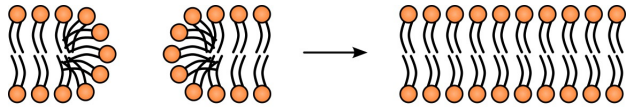
Membrane properties



Vesicle fusion



Genome release

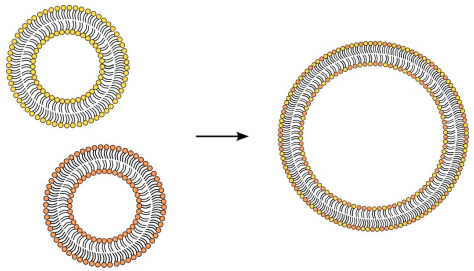


Pore closure

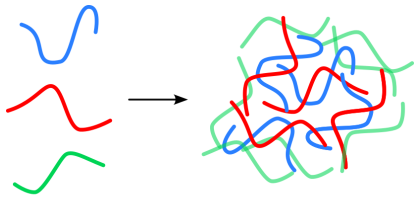


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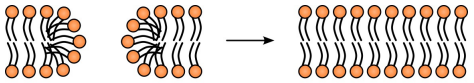
# Non-Equilibrium Simulations



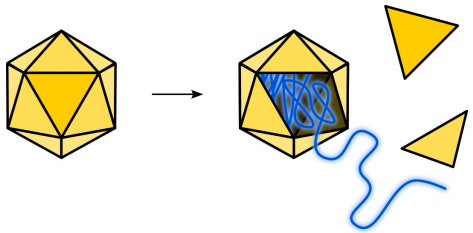
Vesicle fusion



Aggregation



Pore closure



Genome release

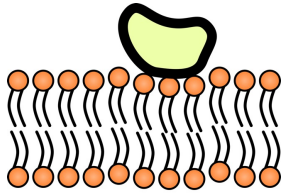
Examples of non-equilibrium simulations

- Multiple starting configurations!
  - To capture a range of possible responses
- Time-dependent observables
  - Fluxes, structural changes, or relaxation times
- External perturbations
  - Drive the system out of equilibrium
  - Temperature or pressure gradients, electric field
- Types:
  - Steady-state: perturbations are constantly applied to study the response under constant conditions
  - Transient: relaxation of a system towards equilibrium state after a sudden perturbation

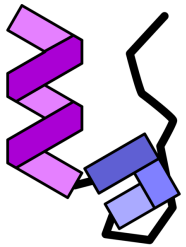


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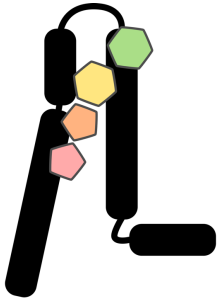
# Equilibrium Simulations



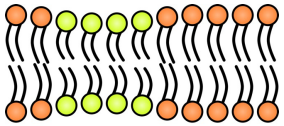
Membrane adsorption



Conformational ensemble



Ligand binding



Membrane properties

Examples of equilibrium simulations

- Single simulation or multiple replicas
  - Start from a well-equilibrated state
- Time-averaged observables
  - Temperature, pressure, …
- System at equilibrium
  - System remains in a steady state as there are no external perturbations
- Thermodynamic properties
  - Calculate enthalpy and free energy



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# Ergodic Hypothesis

- Time averaging is equivalent to ensemble averaging
  - Only if the sampling of system's properties is done over a long enough time period
  - Ensemble average = average over many different configurations
- Importance sampling
  - Over a sufficiently long simulation, the system explores all accessible states

# Molecular Dynamics Simulations

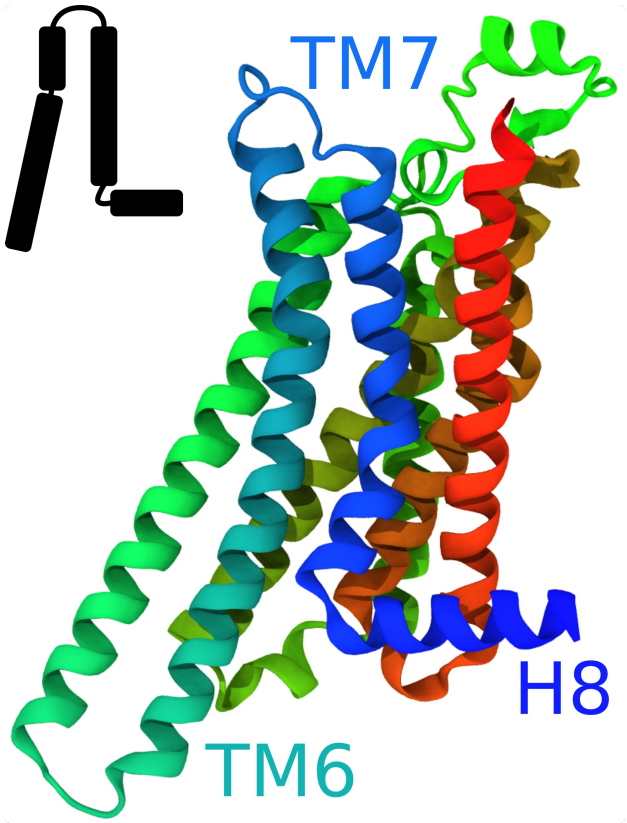
Equilibrium

1. Initial configuration
2. Equilibration
3. Production run
4. Analysis

Non-Equilibrium

1. Multiple initial configurations
2. Production runs
3. Analysis

# G Protein-Coupled Receptor (GPCR)



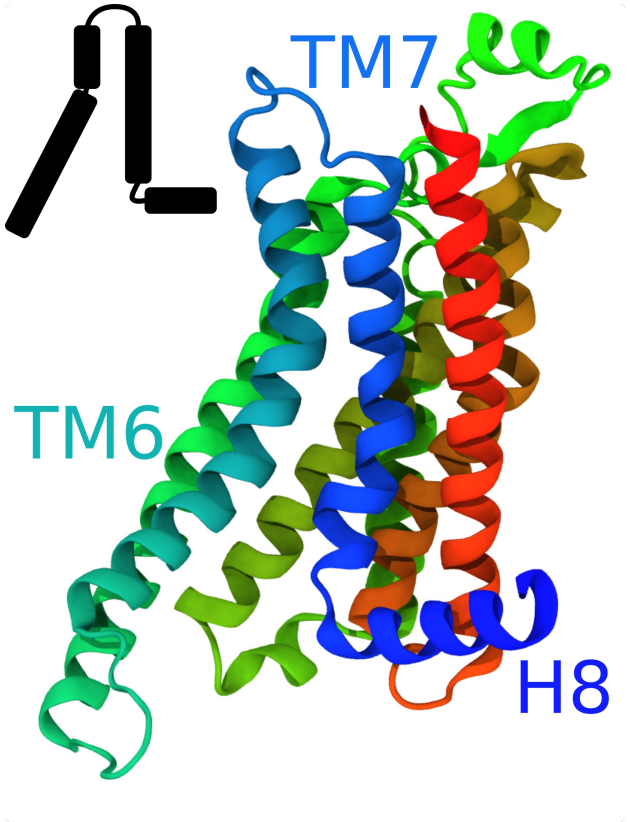

PDB Code: 5MZP

- GPCRs are involved in a wide range of biological processes, including neurotransmission, hormonal regulation, immune response, and sensory perception (e.g., vision, taste, smell)
- Categorized into five main classes (A, B, C, F, and T), with Class A (Rhodopsin-like receptors) being the largest and most studied group
- Approximately 30–40% of all approved drugs target GPCRs
- The estimated 800–1,000 GPCRs represent approximately 3–5% of all human genes



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# G Protein-Coupled Receptor (GPCR)




PDB Code: 6GDG

- GPCRs are involved in a wide range of biological processes, including neurotransmission, hormonal regulation, immune response, and sensory perception (e.g., vision, taste, smell)
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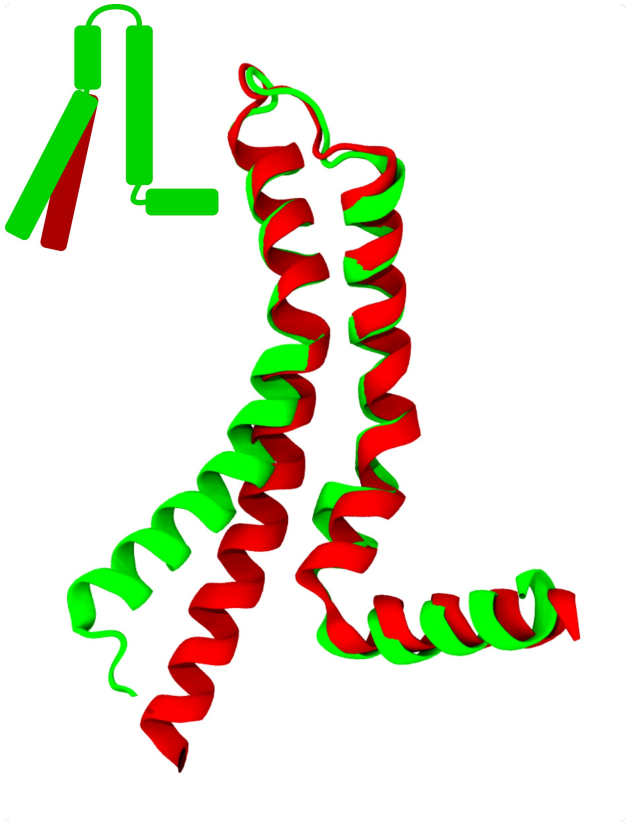
PDB Code: 5MZP and 6GDG

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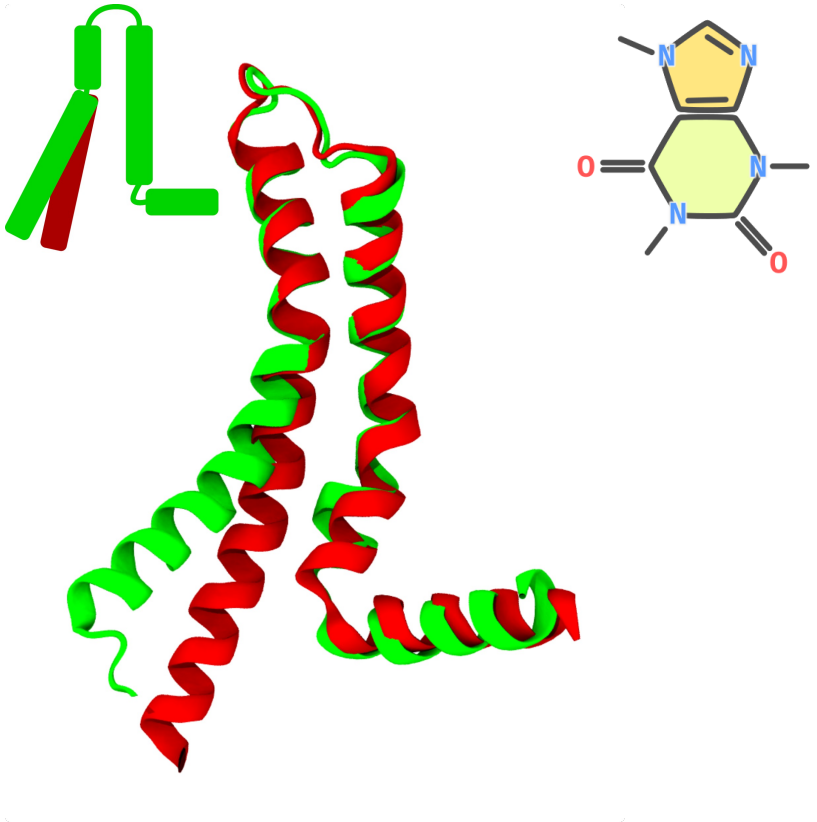
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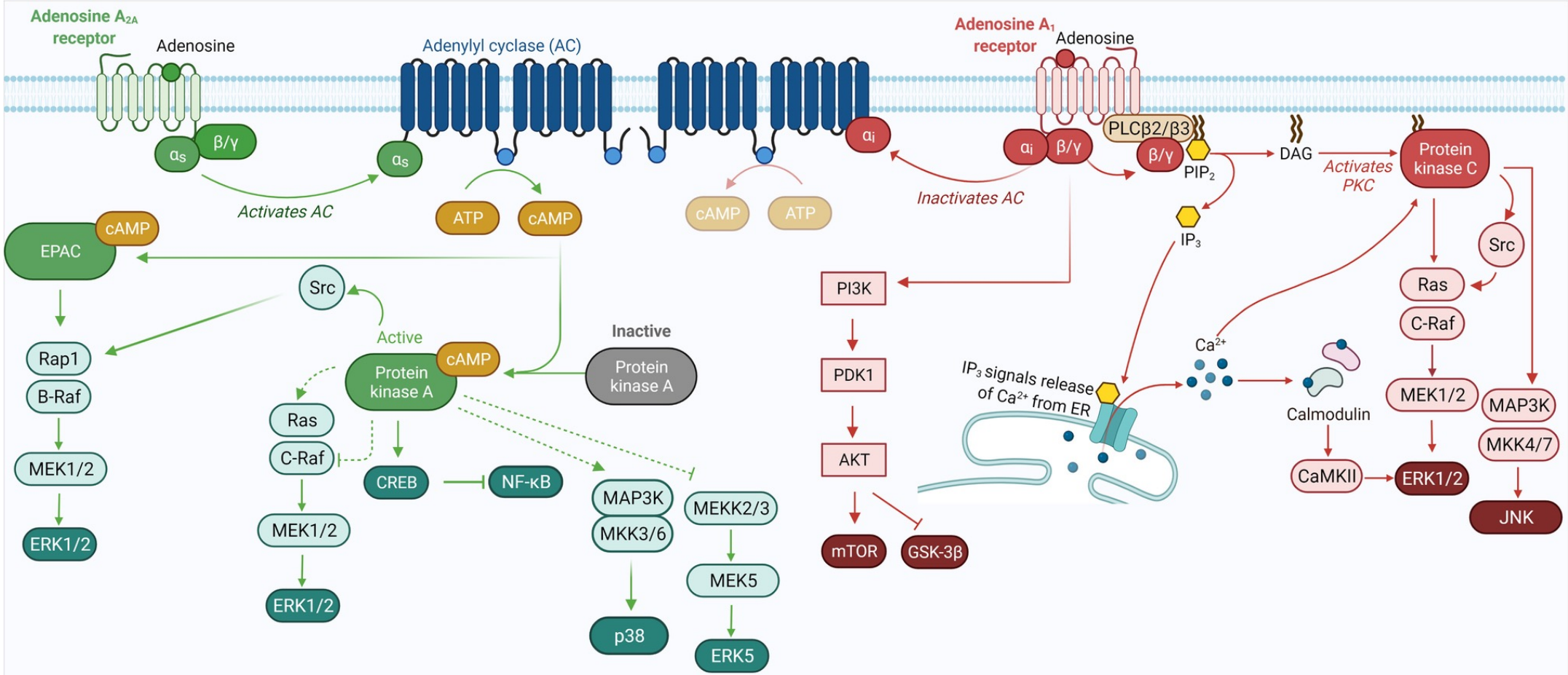

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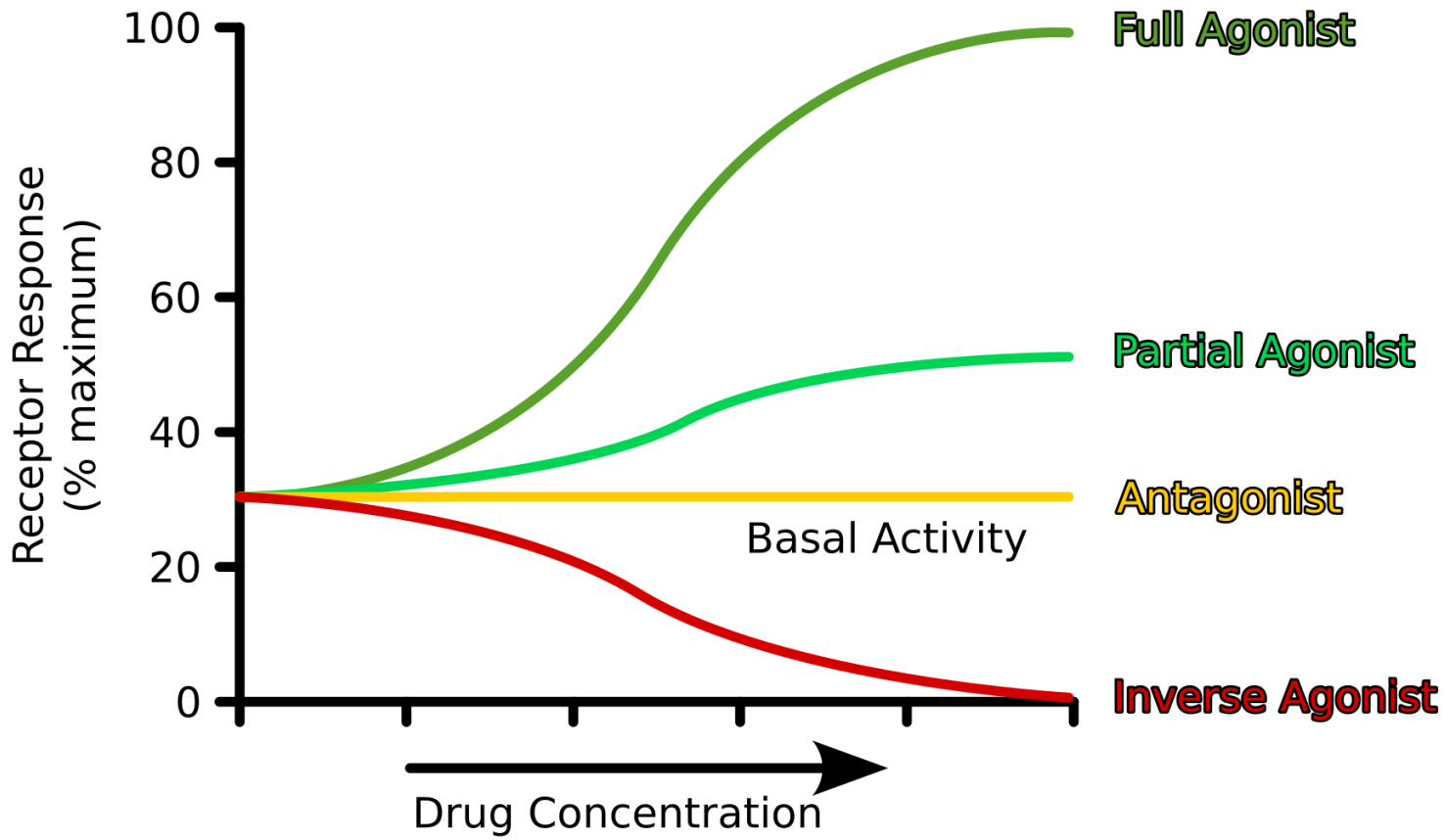
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# Signalling pathways




Trinh, P. N., et al. (2022). Adenosine receptor signalling in Alzheimer's disease. Purinergic signalling, 18(3), 359-381.

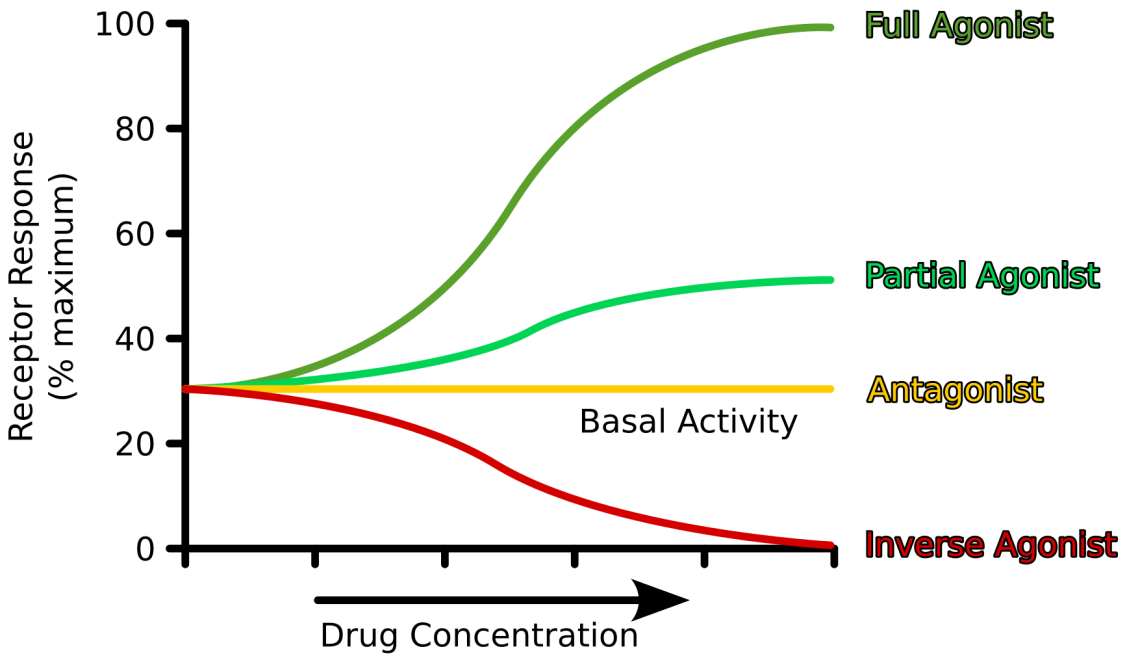
# Dose-Response Curve



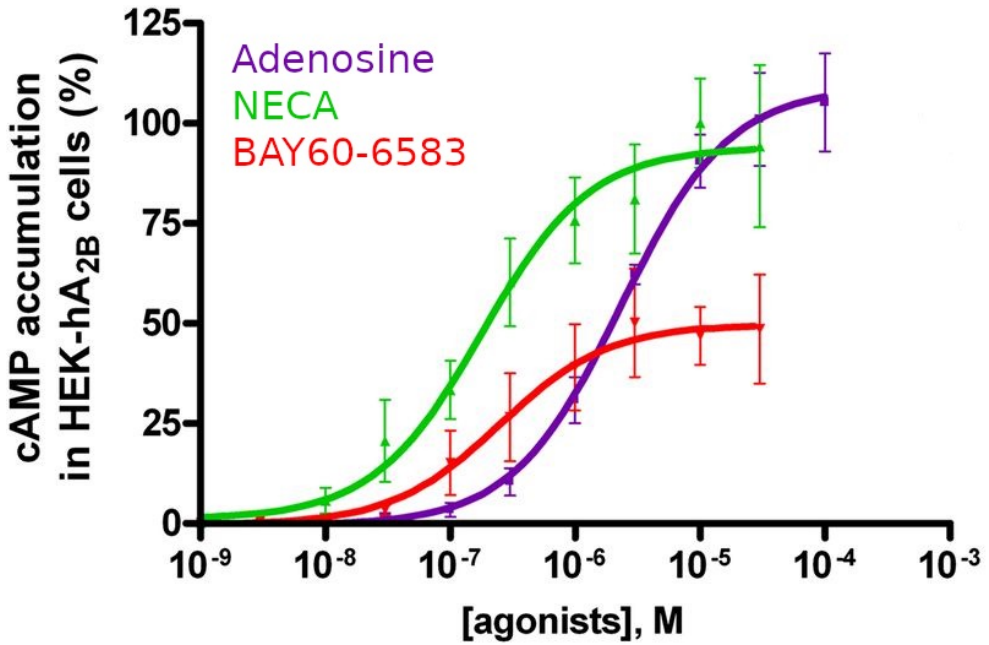



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# Dose-Response Curve




Hinz, S., et al. (2014). Journal of Pharmacology and Experimental Therapeutics, 349(3), 427-436.






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# Free Energy

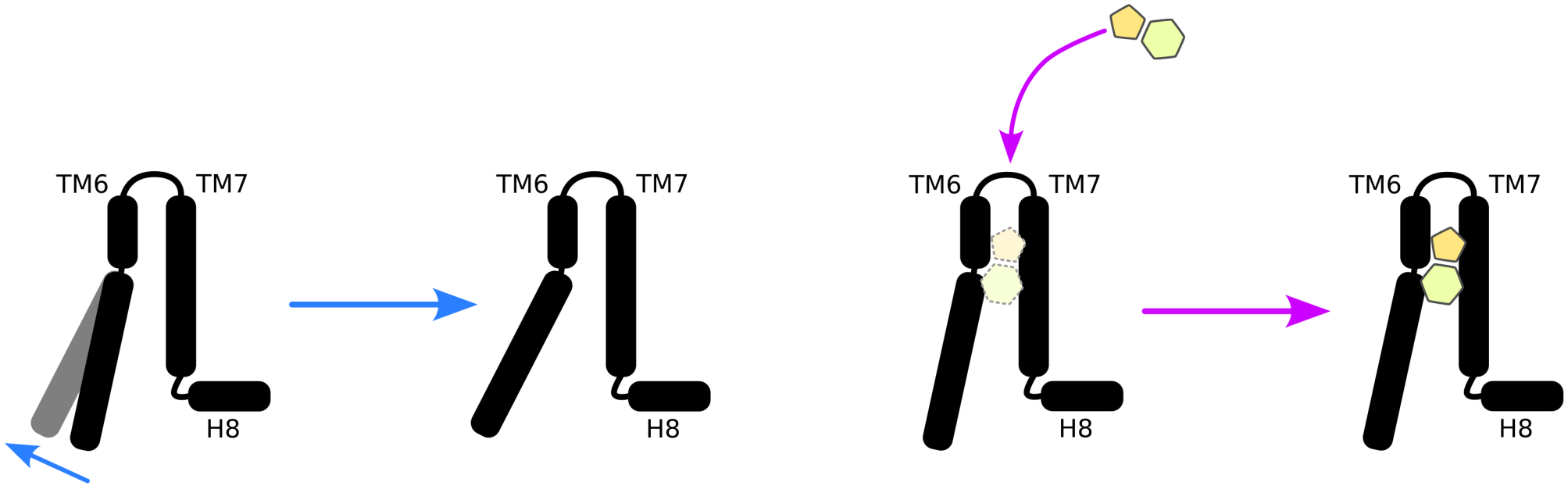
- Is a state function
  - Depends only on the initial and final states, not on the path taken
- Explains how molecules populate (distribute among) different energy states

- The change in Gibbs free energy, ΔG, for a process is given by:
  - $\Delta G = \Delta H - T\Delta S$
    **ΔG**: Change in Gibbs Free Energy
    **ΔH:** Change in enthalpy
    **ΔS**: Change in entropy
    Determined under constant temperature and pressure
  - A negative ΔG indicates that the process is spontaneous
  - Spontaneity does not imply speed; it only indicates thermodynamic favorability

# Collective Variable (CV)

- A parameter that describes the progress of a process
  - Reduces the complexity of many-particle systems to a few key degrees of freedom
  - Must accurately represent the process or transition of interest
  - Selecting a CV can be non-trivial and often requires domain expertise
- Used to simplify the analysis of a trajectory or for enhanced sampling
  - In enhanced sampling, CVs facilitate efficient sampling of rare events or slow processes
- Examples of common collective variables:
  - Distance between atoms or molecules
  - Angles and dihedral angles
  - Coordination numbers
  - Radii of gyration

# Collective Variable (CV)




## Receptor activation

The outward swing of TM6 in a receptor can be described as either a distance or an angle

## Ligand binding

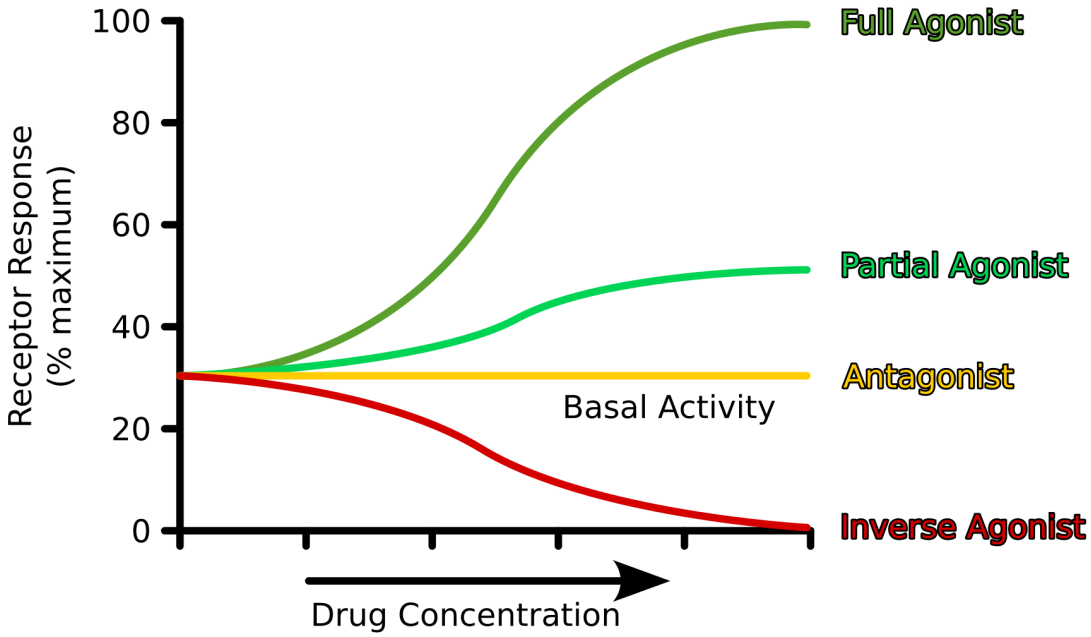
Distance of a ligand from a receptor binding pocket



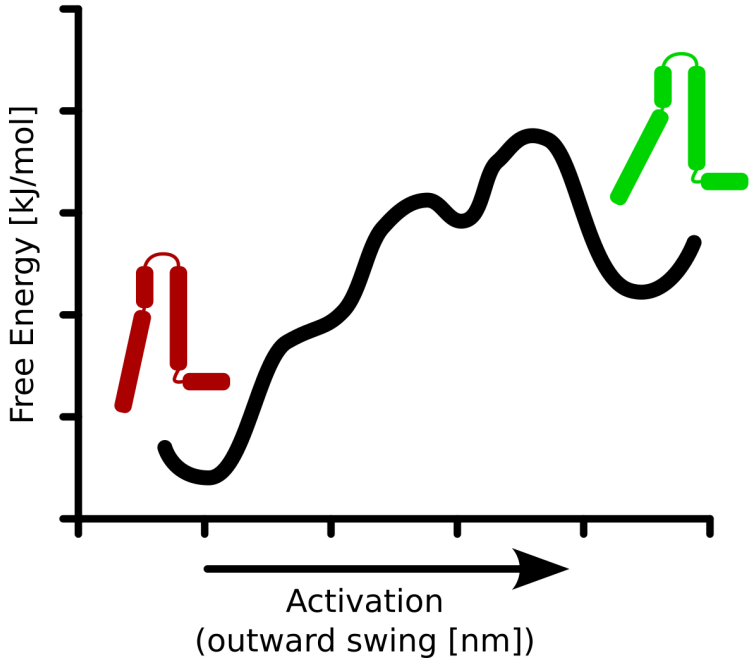
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# Populations and Free Energy

Dose-response curve




Free energy

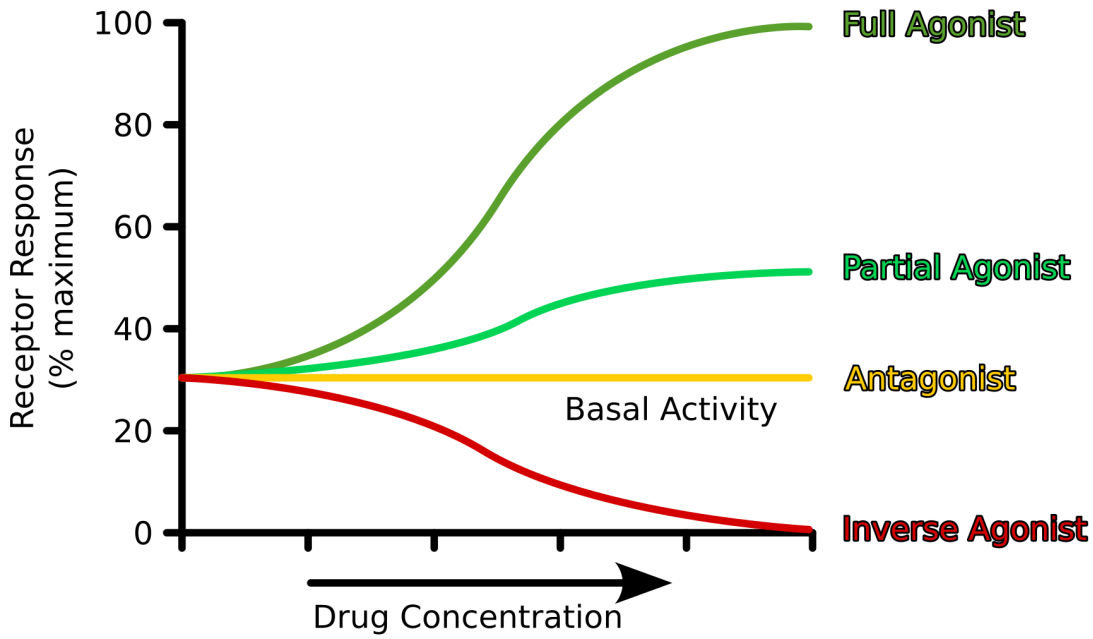




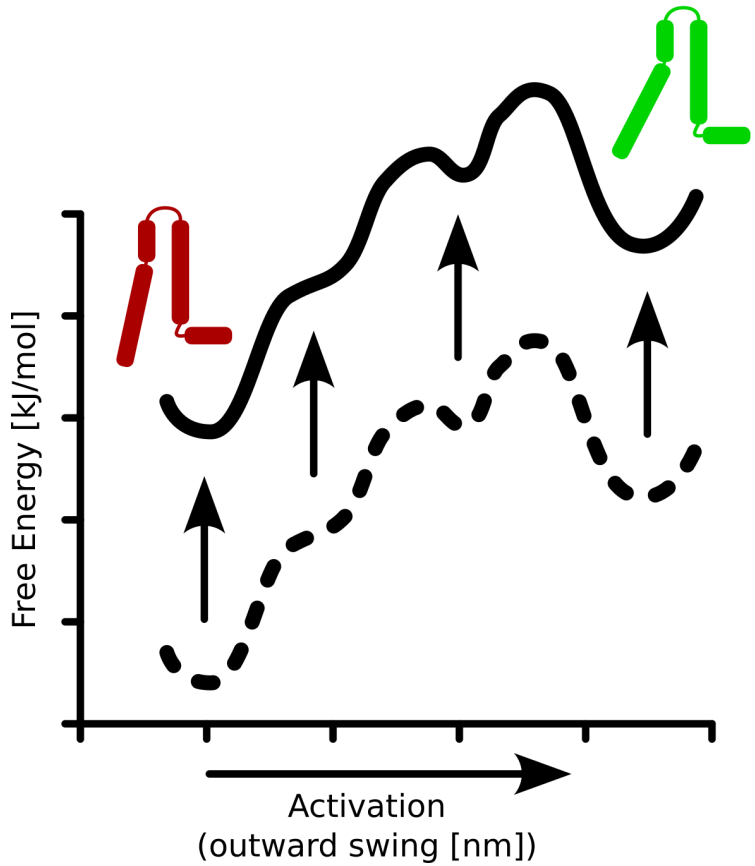

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# Populations and Free Energy

## Dose-response curve




## Free energy

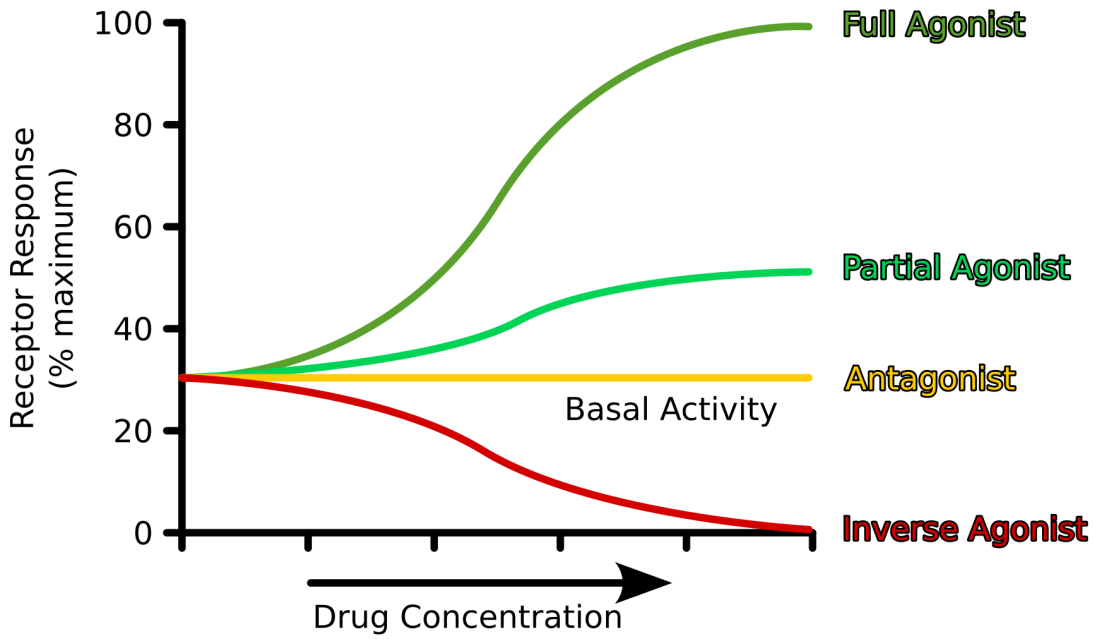




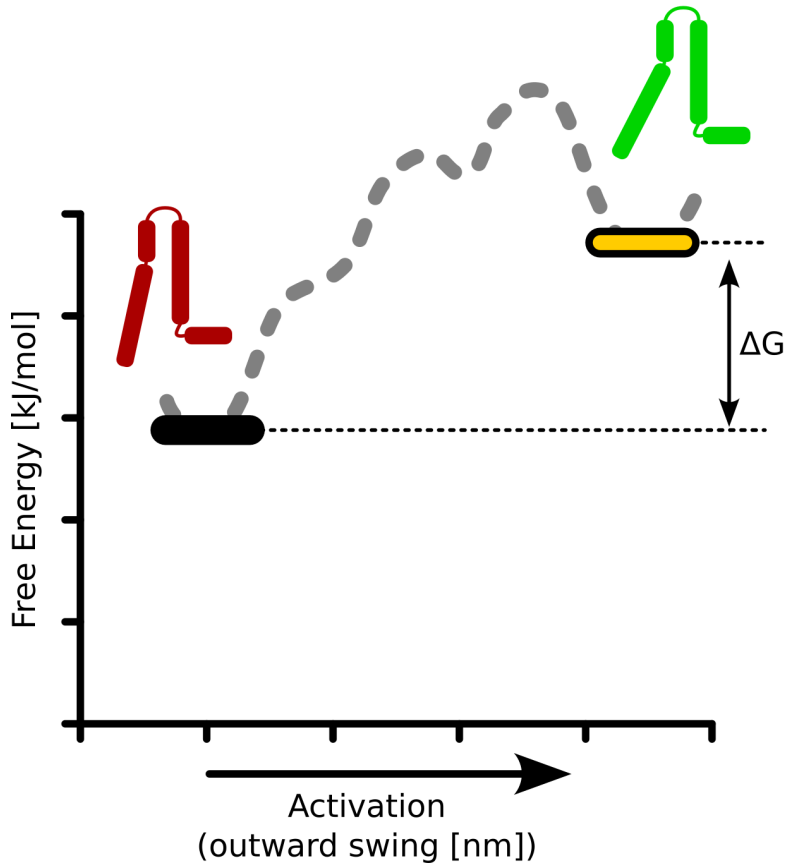

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# Populations and Free Energy

## Dose-response curve




## Free energy

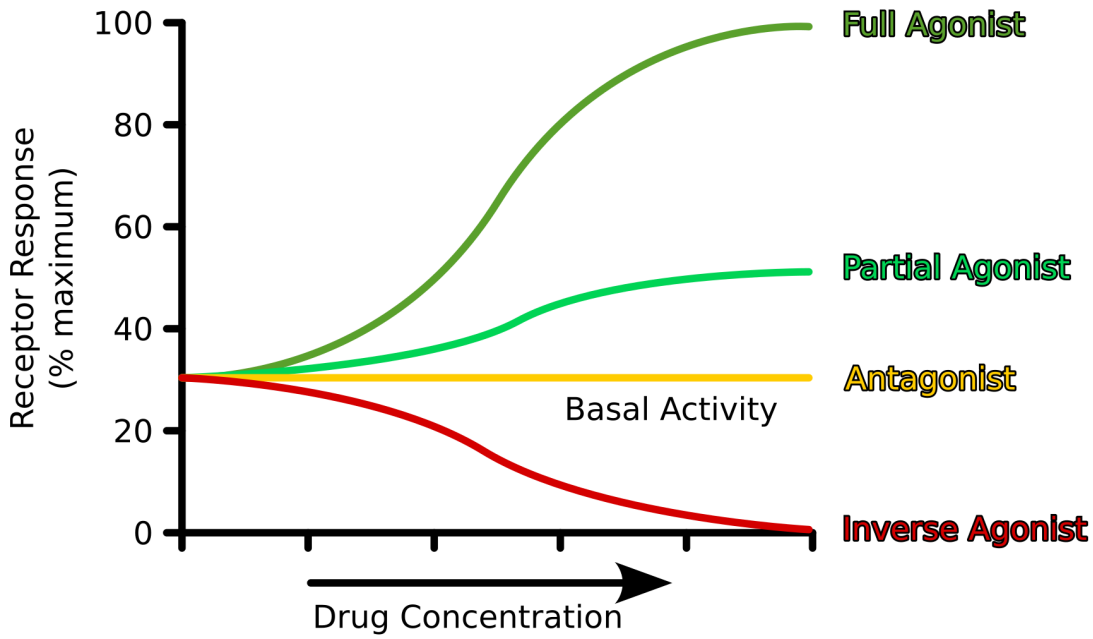




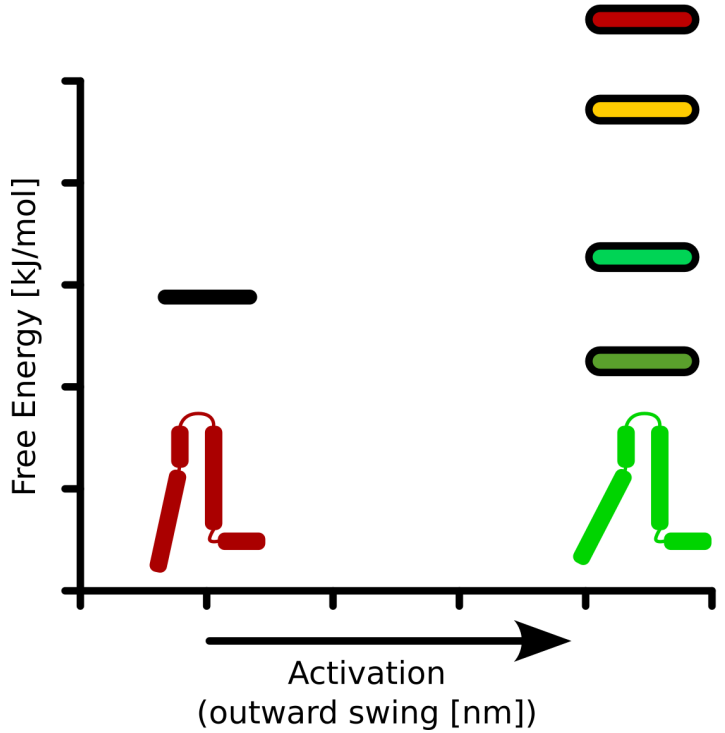

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# Populations and Free Energy

## Dose-response curve




## Free energy

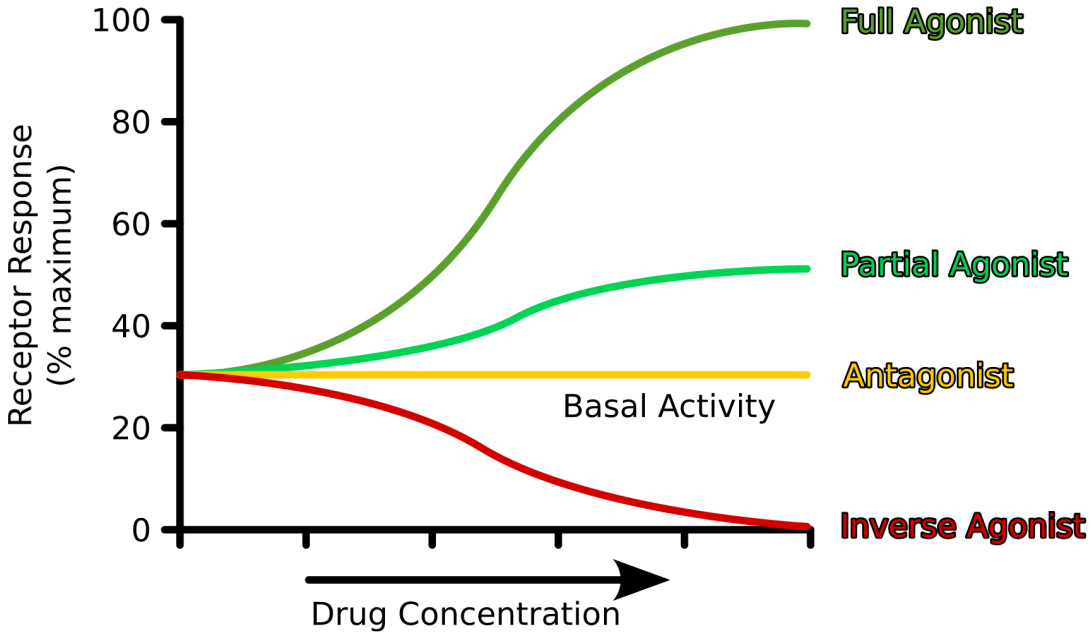




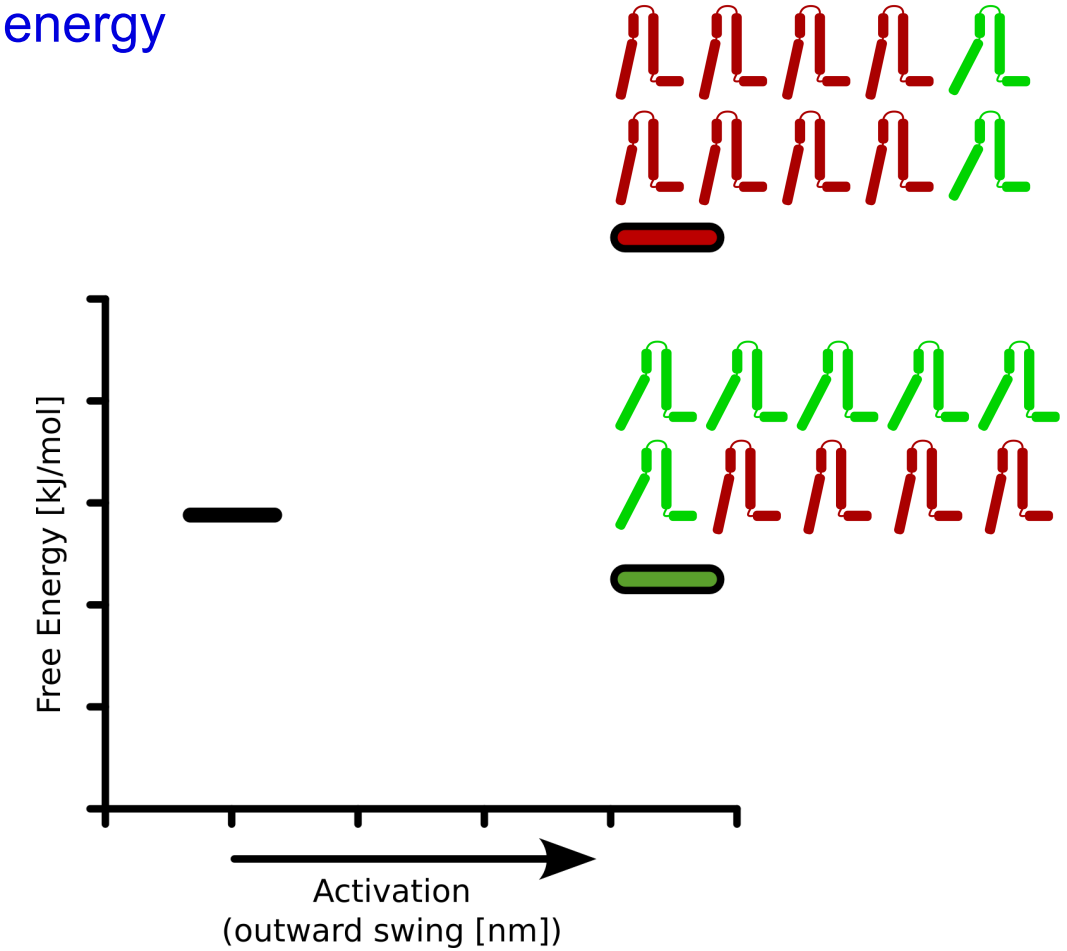

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# Populations and Free Energy

## Dose-response curve



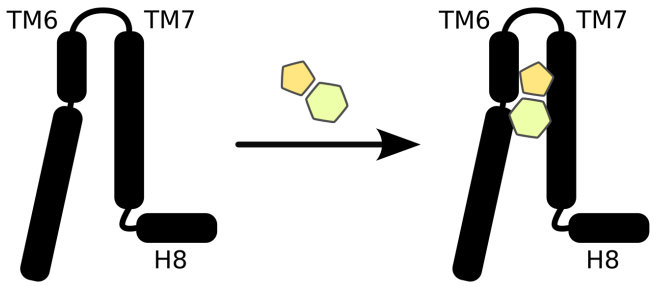

## Free energy






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# Connecting Simulations and Experiment



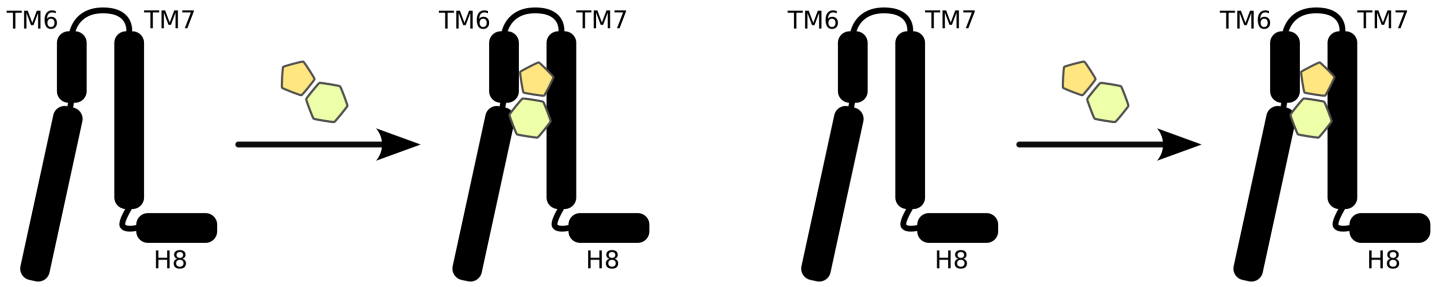

Dissociation constant ($K_D$)

Experimentally determined using, e.g., surface plasmon resonance (SPR), isothermal titration calorimetry (ITC), or fluorescence spectroscopy.



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# Connecting Simulations and Experiment



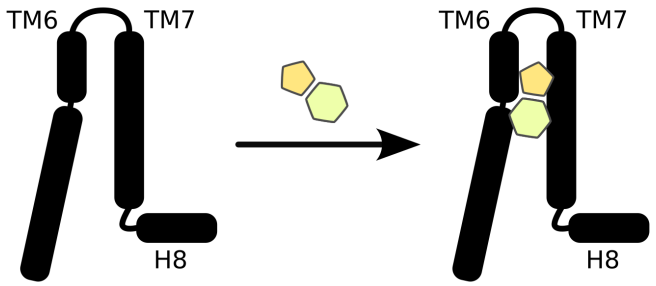

Experimentally determined using, e.g., surface plasmon resonance (SPR), isothermal titration calorimetry (ITC), or fluorescence spectroscopy.

Dissociation constant $K_D$ is calculated from the concentrations of free ligand, receptor, and ligand-receptor complex, representing the equilibrium populations of different states.



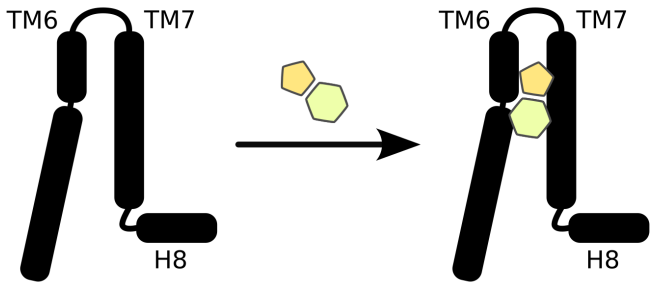
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# Connecting Simulations and Experiment



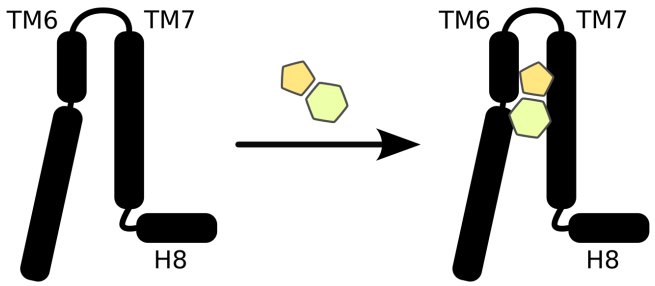

Dissociation constant ($K_D$)

Experimentally determined using, e.g., surface plasmon resonance (SPR), isothermal titration calorimetry (ITC), or fluorescence spectroscopy




$$K_D = \frac{[L][R]}{[LR]}$$

Dissociation constant $K_D$ is calculated from the concentrations of free ligand, receptor, and ligand-receptor complex, representing the equilibrium populations of different states.

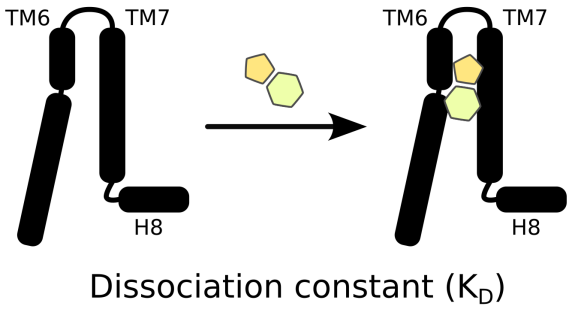



$$\Delta G^\circ = RT \ln K_D$$

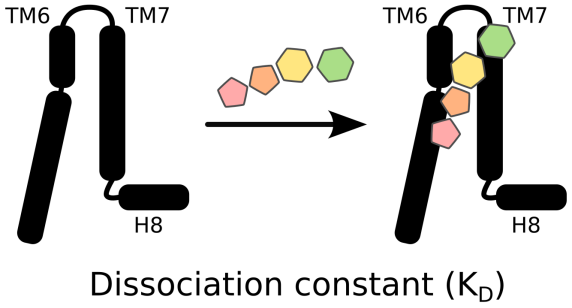
Relationship between the dissociation constant $K_D$ and standard Gibbs free energy $\Delta G^\circ$.



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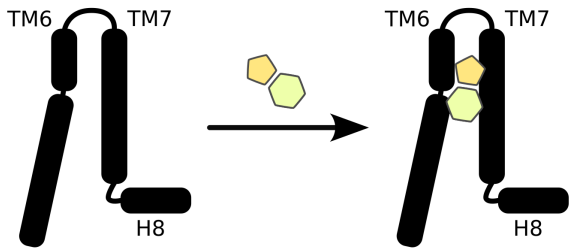


Dissociation constant ($K_D$)

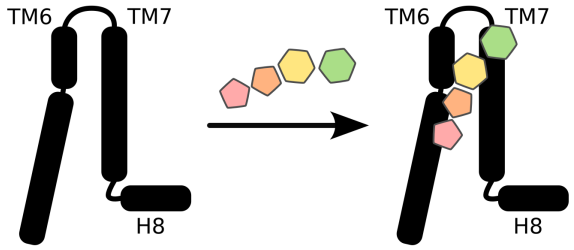


Dissociation constant ($K_D$)

- Weakly binding ligand
  - Higher $K_D$
  - Less negative free energy
- Strongly binding ligand
  - Lower $K_D$
  - More negative free energy
- Commonly reported as pKD
  - $pK_D = -\log_{10} K_D$
  - Higher $pK_D$ indicate stronger binding
  - If $K_D$ = 1 µM, then $pK_D$ = 6
  - If $K_D$ = 1 nM, then $pK_D$ = 9




Dissociation constant ($K_D$)

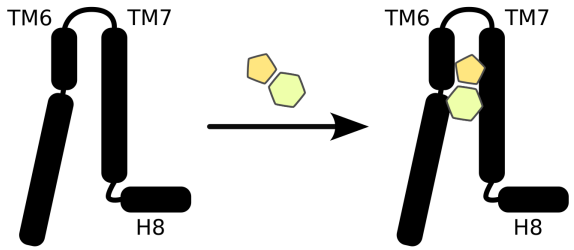



Dissociation constant ($K_D$)

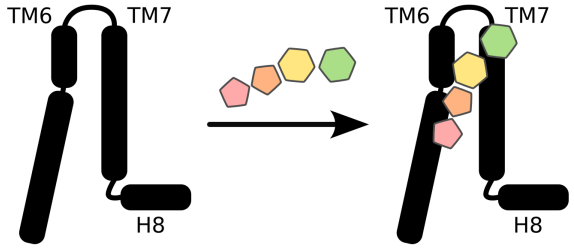




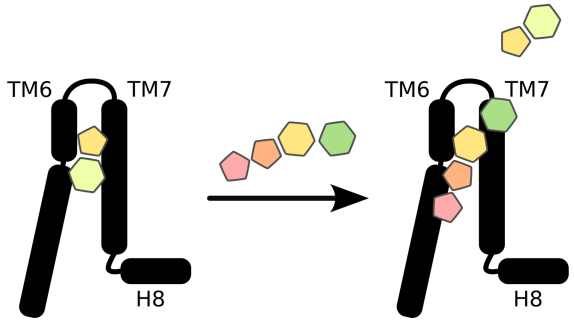
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Dissociation constant ($K_D$)




Dissociation constant ($K_D$)

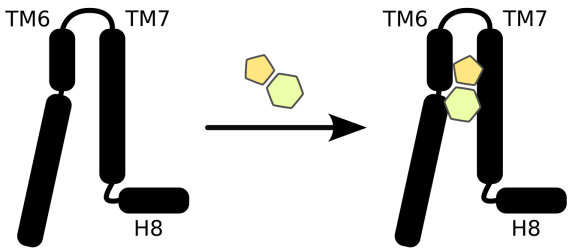



Inhibition constant ($K_I$)

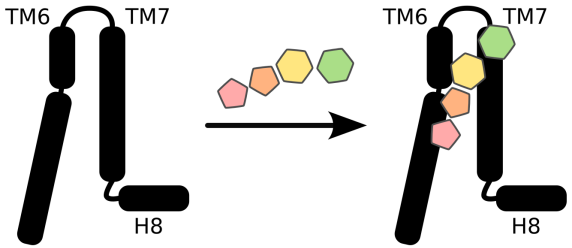
- Inhibition constant, $K_I$
  - Binding affinity via ligand displacement
  - Quantifies how effectively a molecule can prevent a ligand with known affinity from binding



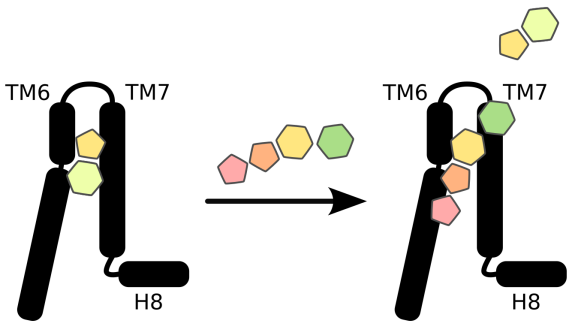
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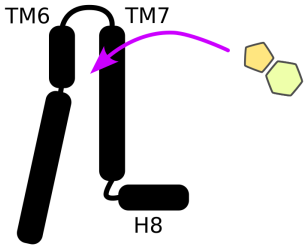

Dissociation constant ($K_D$)



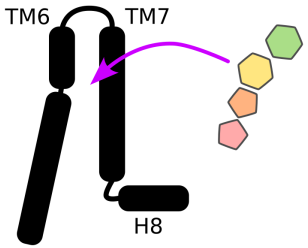

Dissociation constant ($K_D$)



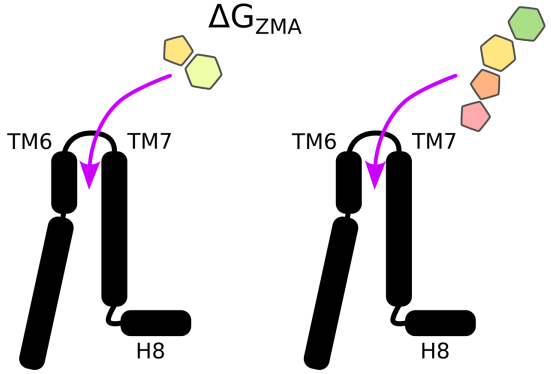

Inhibition constant ($K_I$)




$\Delta G_{CFF}$




$\Delta G_{ZMA}$



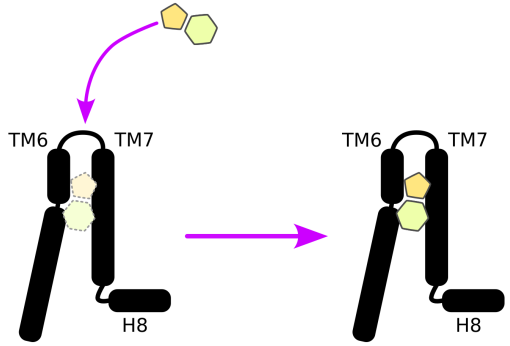

$\Delta\Delta G = \Delta G_{ZMA} - \Delta G_{CFF}$



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# Practical Example 1

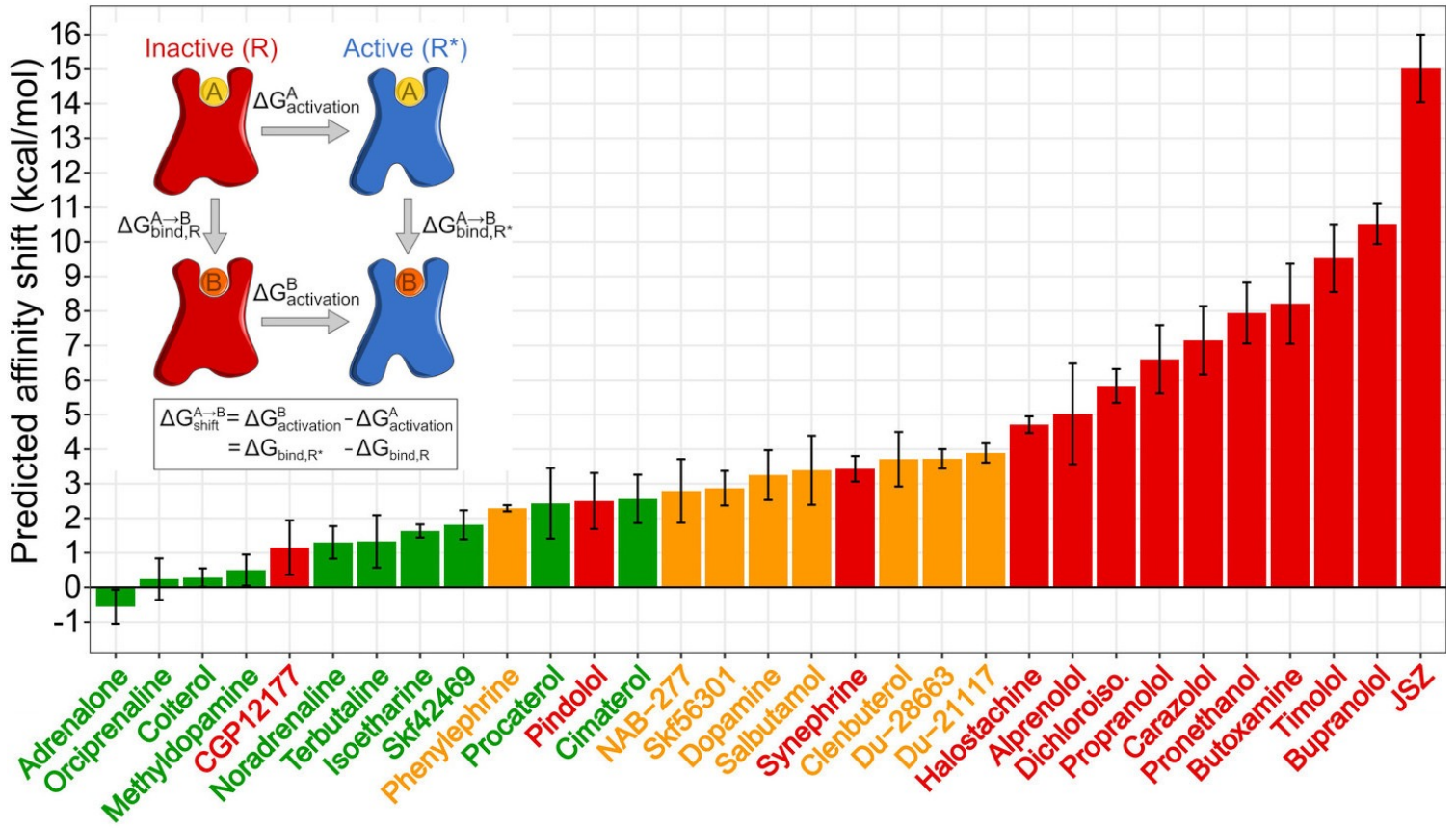
## Differences in ligand affinities for a receptor in its active versus inactive conformation




The distance of a ligand from a receptor binding pocket can be used as a collective variable to calculate the free energy of binding*.

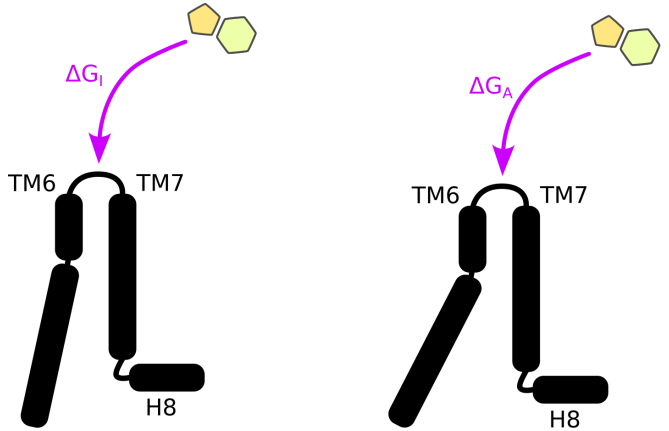
* If all possible poses, orientations, and conformations are not sampled, the result will **NOT** represent the binding free energy but rather the free energy along the chosen coordinate.

Panel, N., et al. (2023). Angewandte Chemie International Edition, 62(22), e202218959.

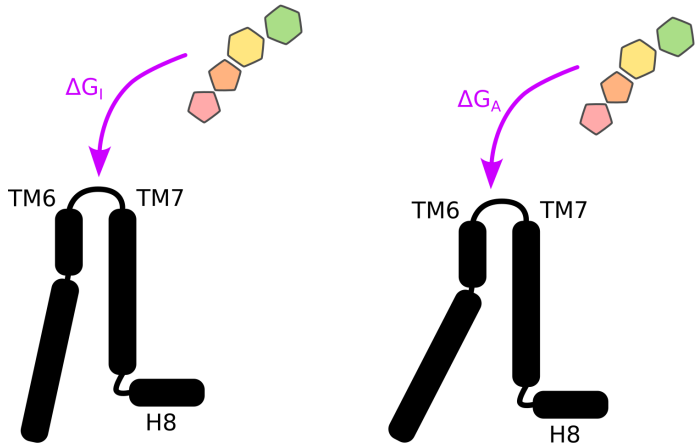





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$$\Delta\Delta G = \Delta G_A - \Delta G_I$$



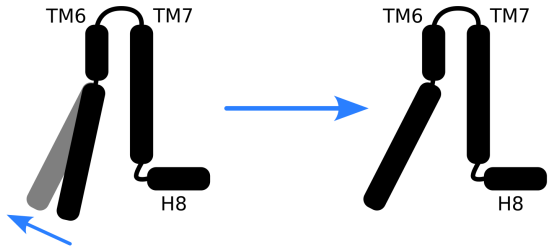

$$\Delta\Delta G = \Delta G_A - \Delta G_I$$



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# Practical Example 2

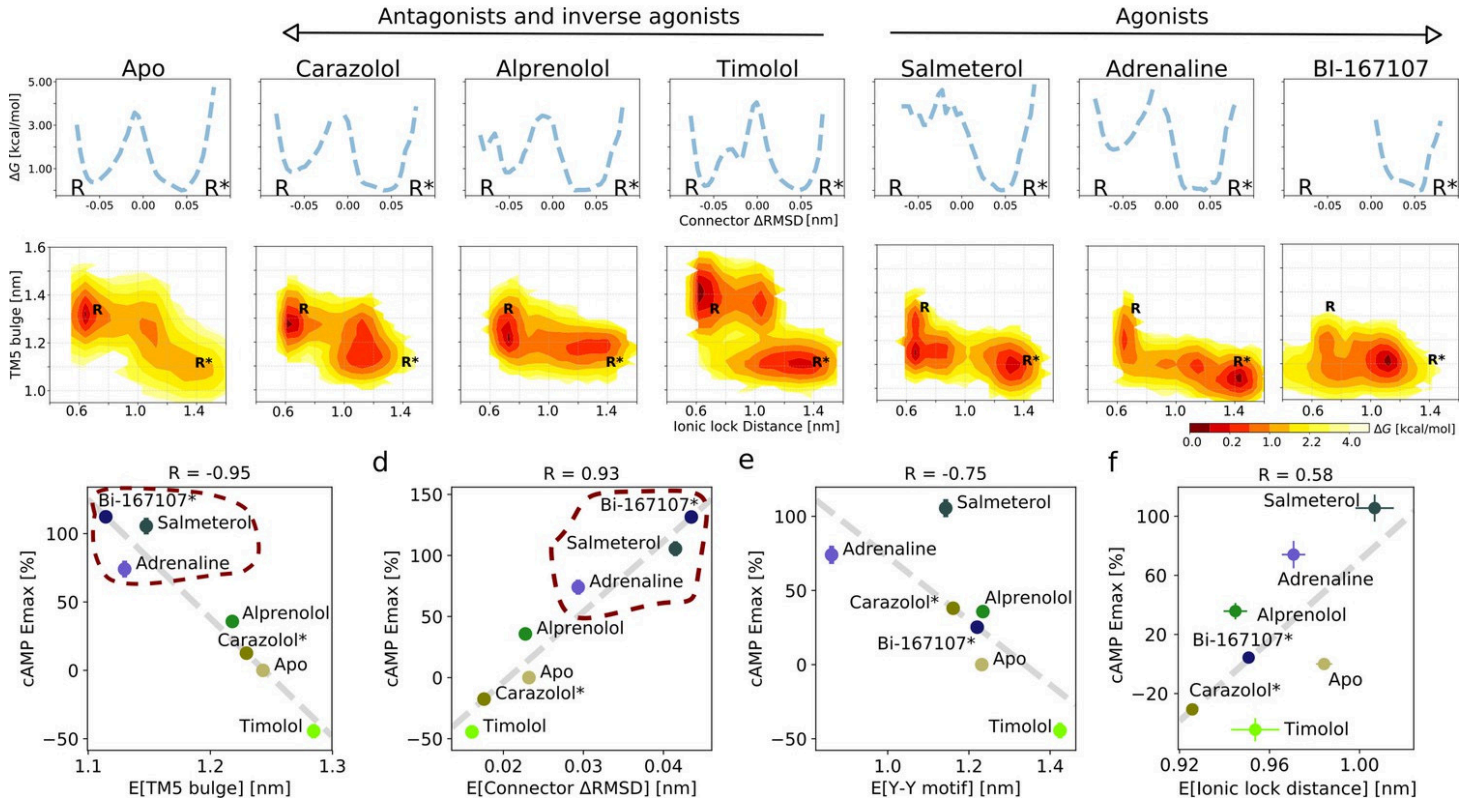
## Ligand-induced change in relative populations of receptors in active and inactive conformations




The swing* of TM6 is used to describe the receptor activation.

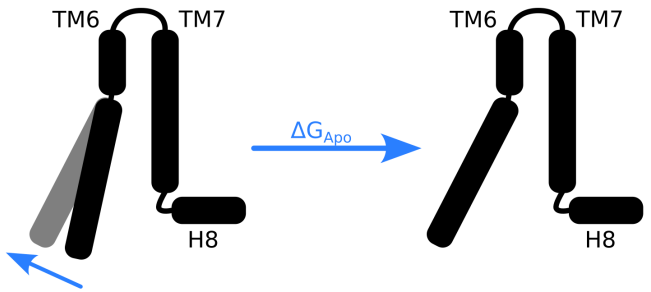
* Simple collective variables are often insufficient for accurately capturing protein conformational changes using enhanced sampling methods.

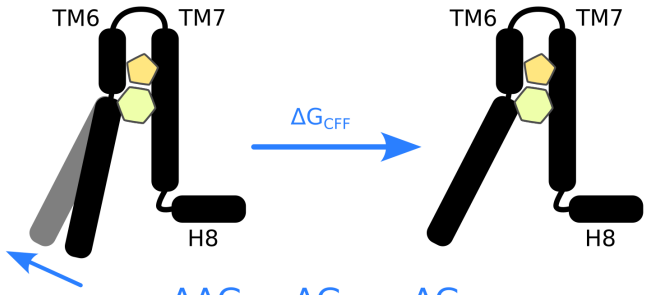
Fleetwood, O., et al. (2021). eLife, 10, e60715



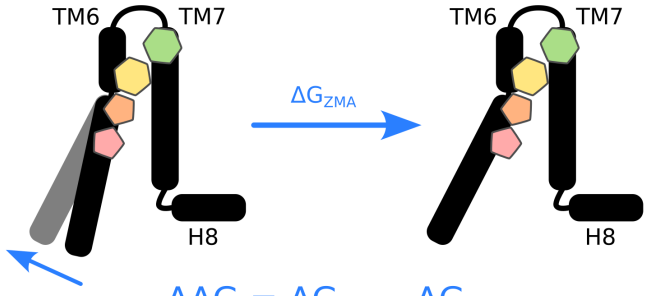



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$$\Delta\Delta G = \Delta G_{CFF} - \Delta G_{Apo}$$

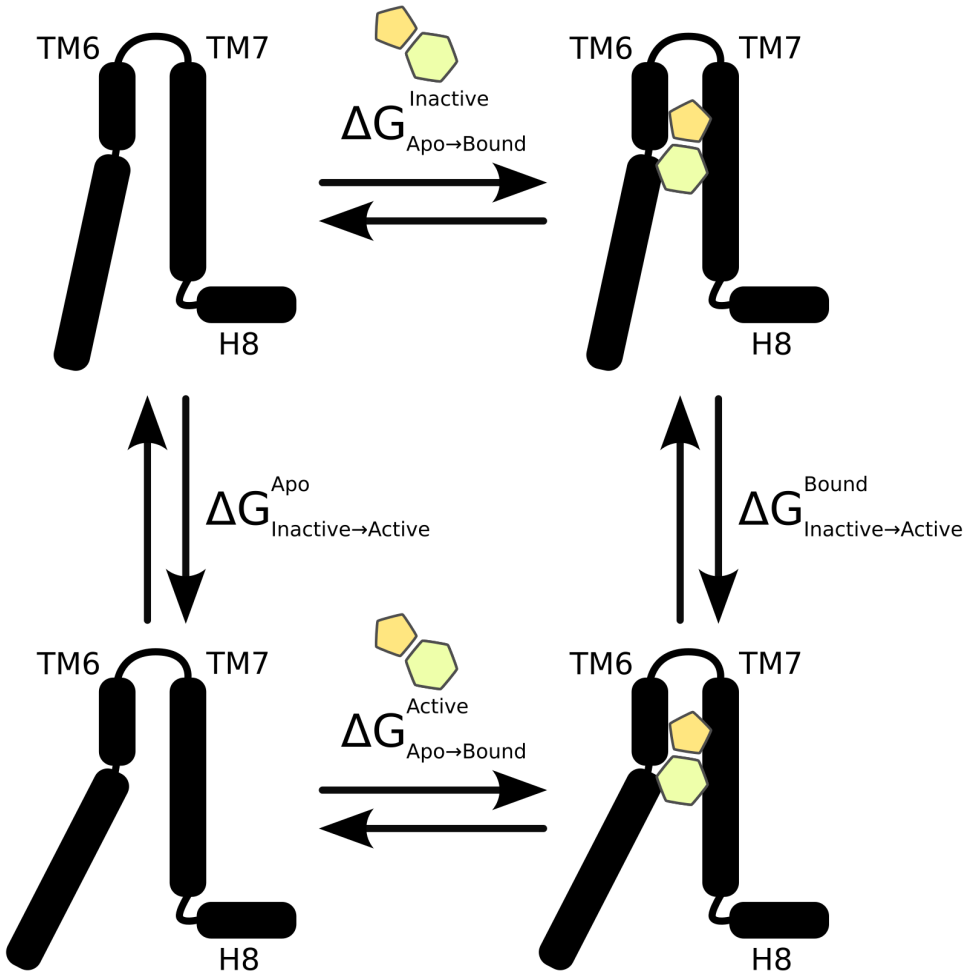



$$\Delta\Delta G = \Delta G_{ZMA} - \Delta G_{Apo}$$



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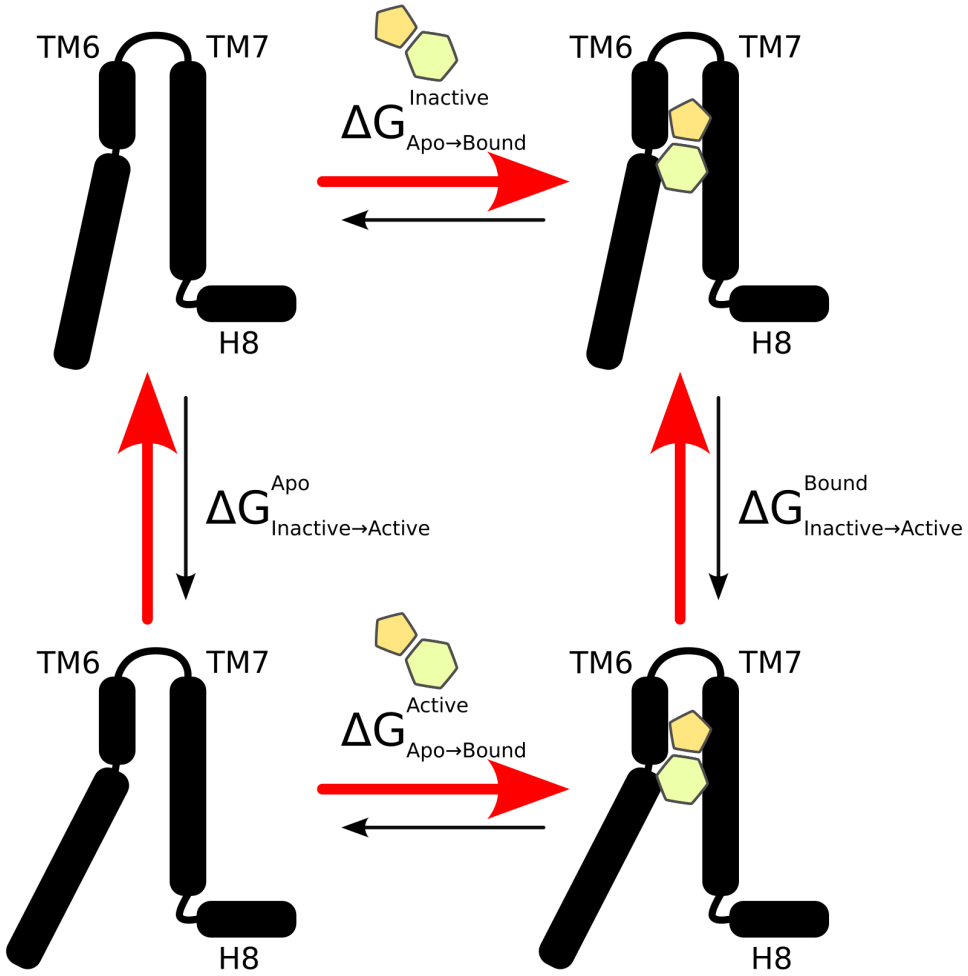
# Thermodynamic Cycle






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# Thermodynamic Cycle



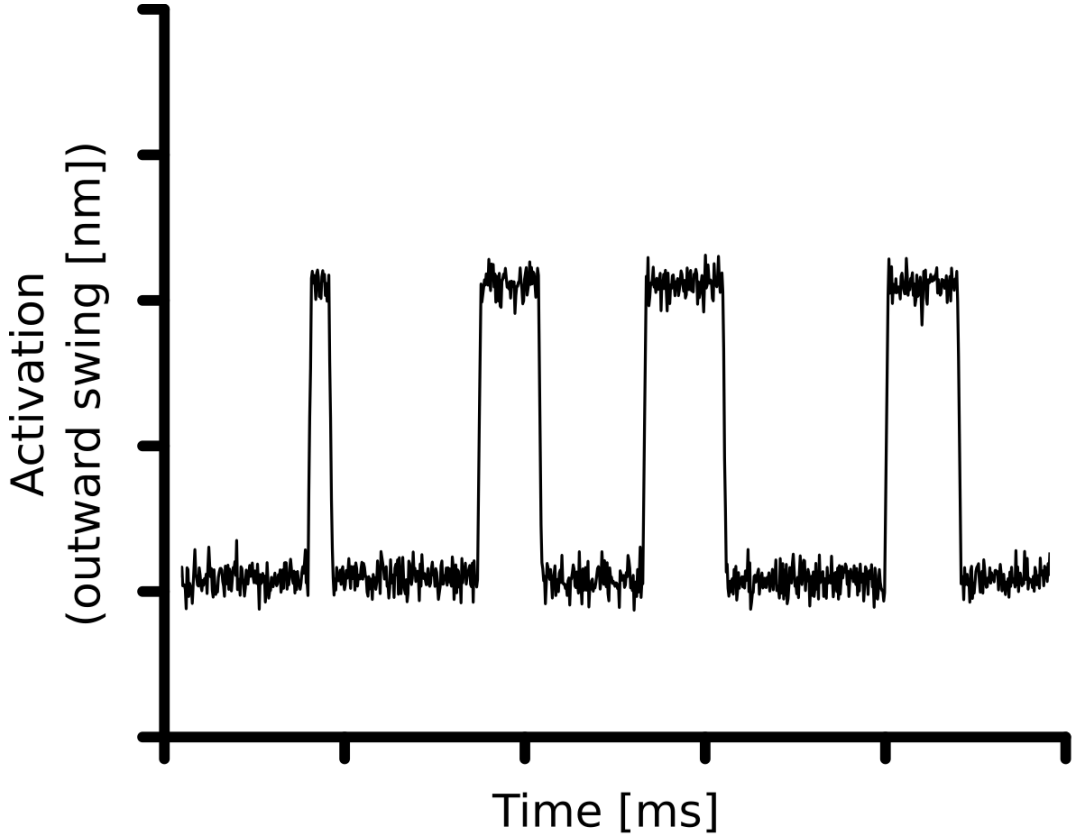


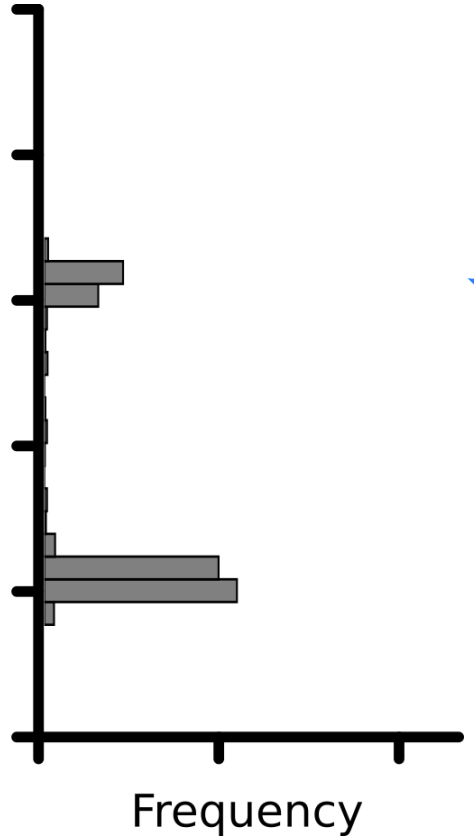

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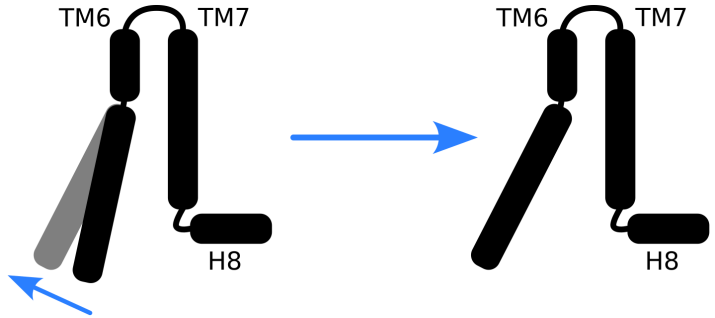
# Ergodic Hypothesis

- Time averaging is equivalent to ensemble averaging
  - Only if the sampling of system's properties is done over a long enough time period
  - Ensemble average = average over many different configurations
- Importance sampling
  - Over a sufficiently long simulation, the system explores all accessible states

# Sampling of Receptor Conformations






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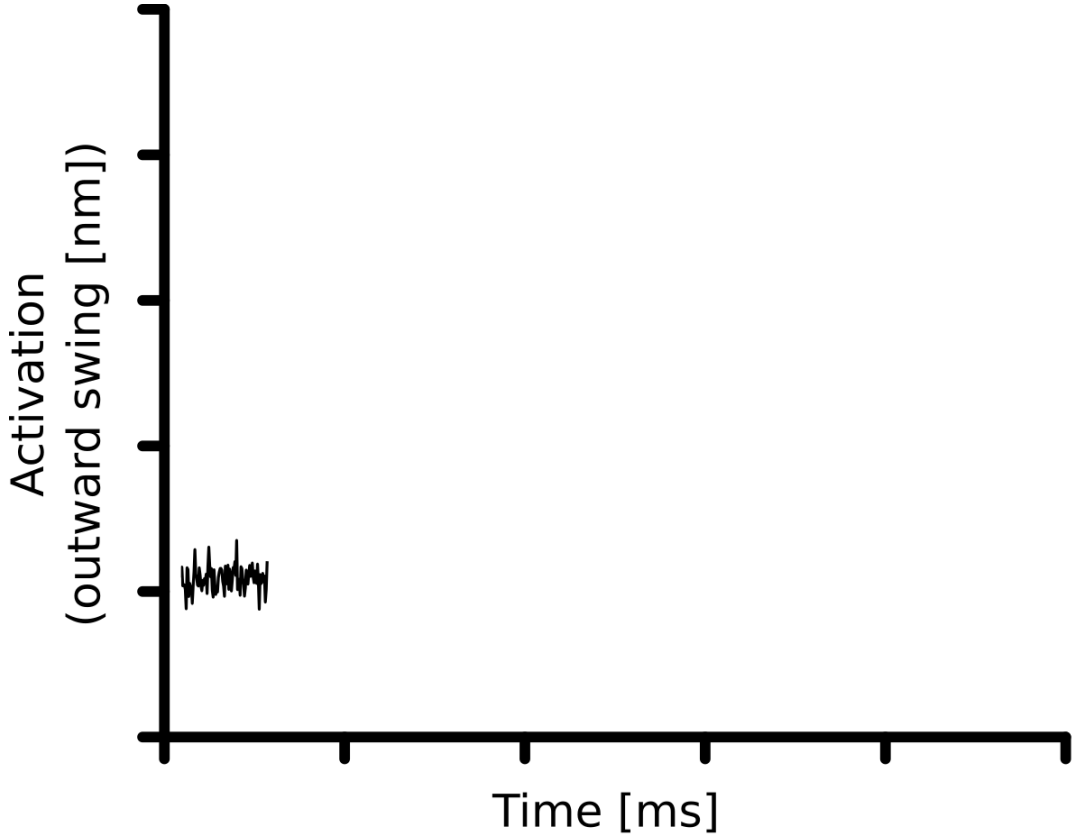
# Convergence Tests

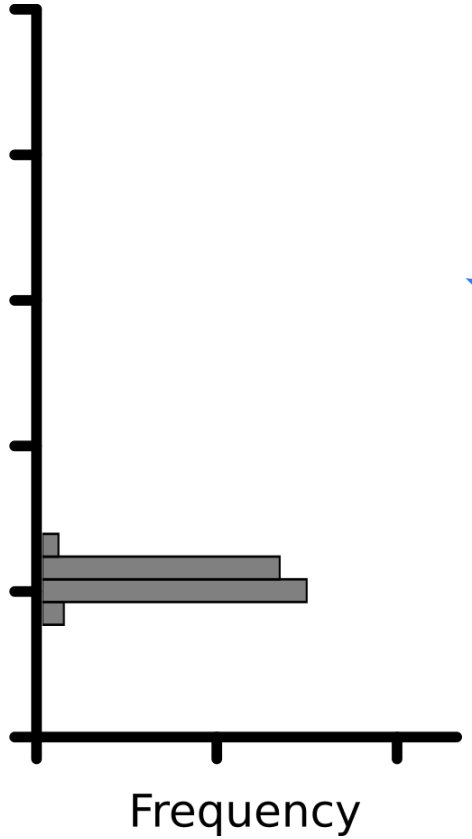
- Increase simulation length
  - Monitor if time-averaged quantities change
- Block averaging
  - Divide a trajectory into blocks of equal length and calculate averages for each block
  - Stratification / strata
- Multiple starting conditions
  - Simulations should yield similar averages
- Check for symmetries

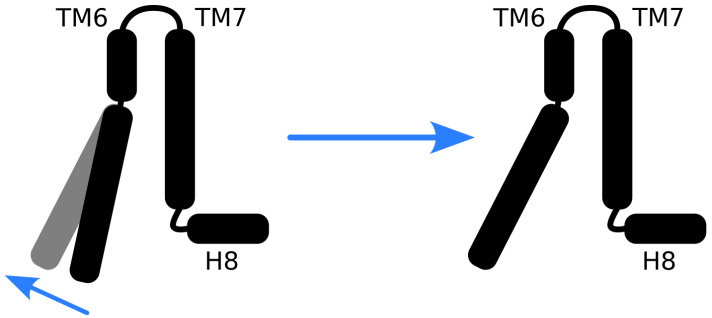
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# Sampling of Receptor Conformations



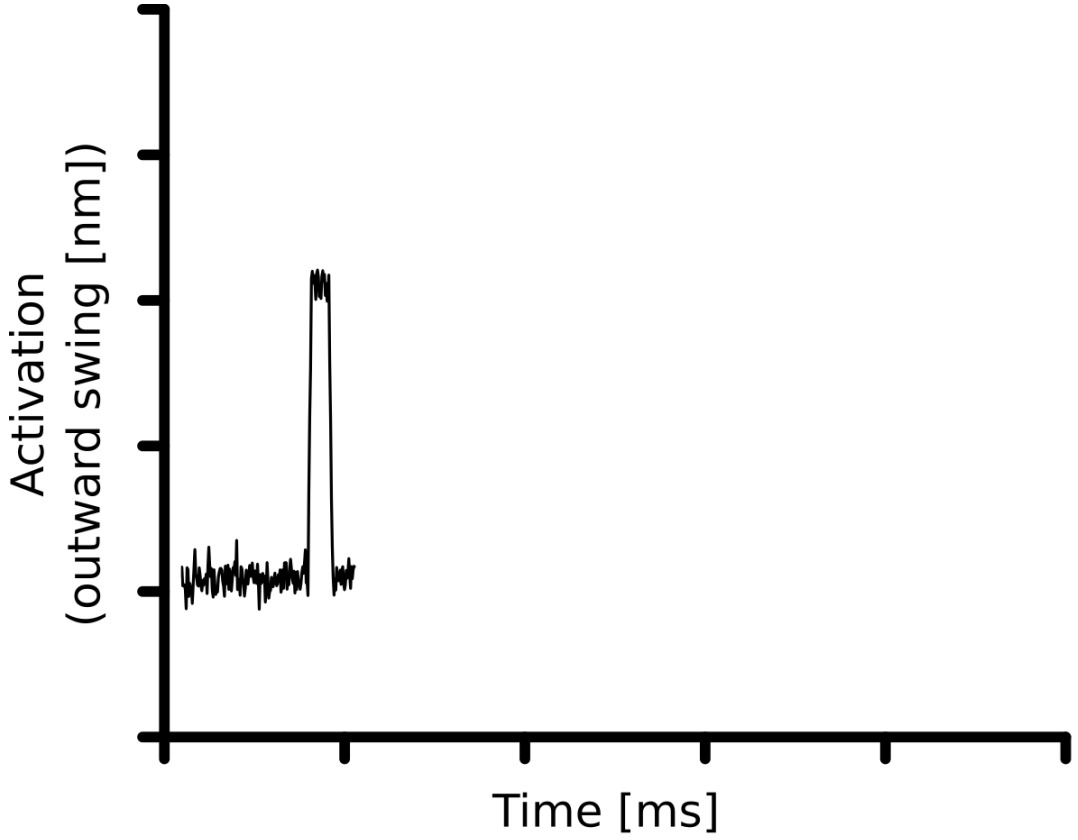



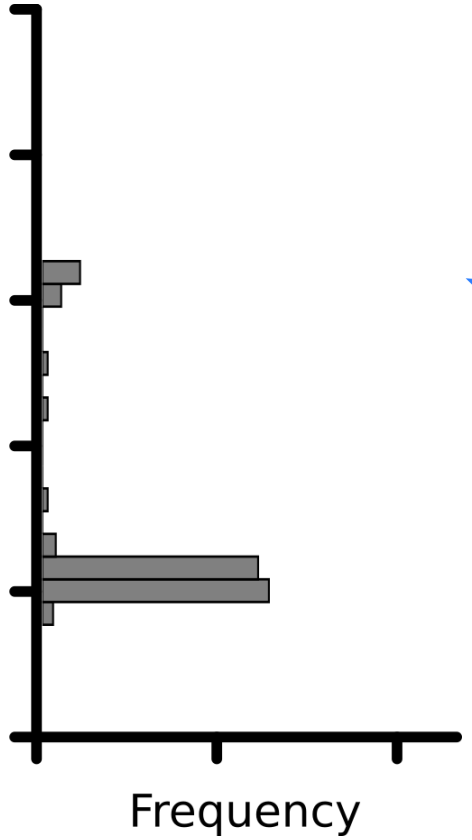



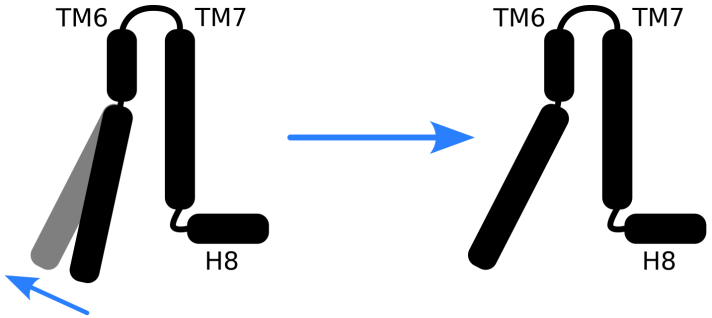



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# Sampling of Receptor Conformations



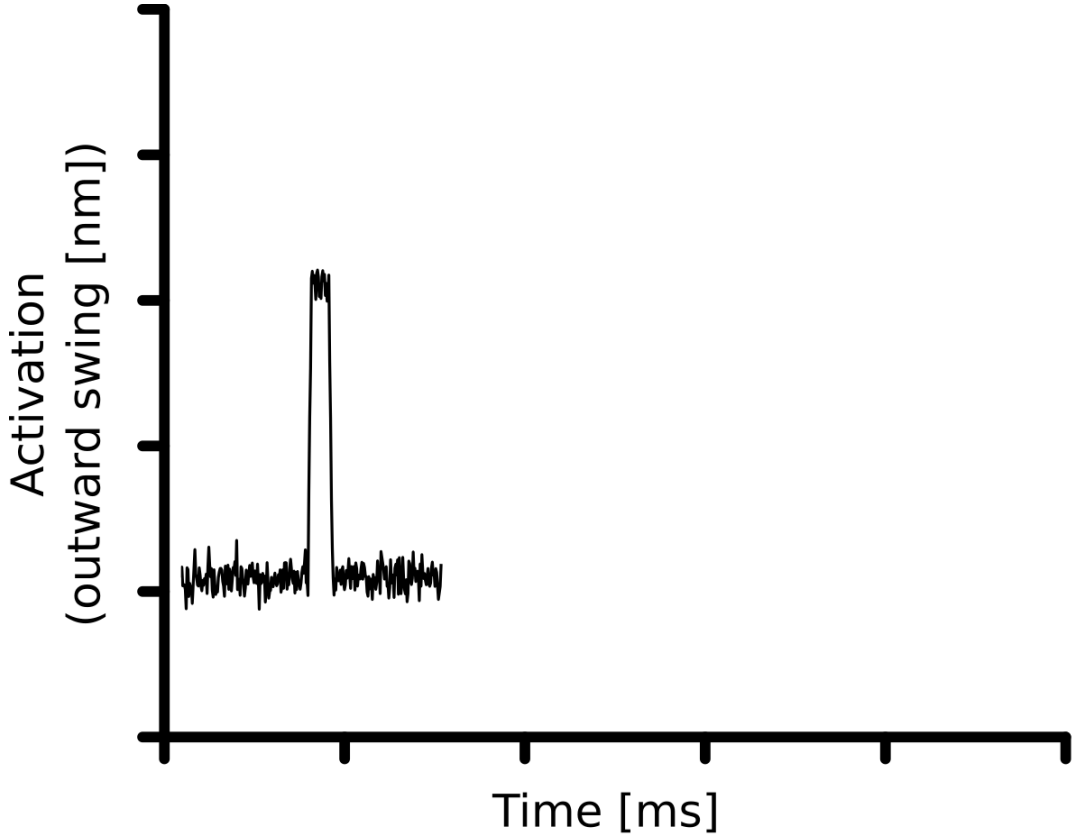



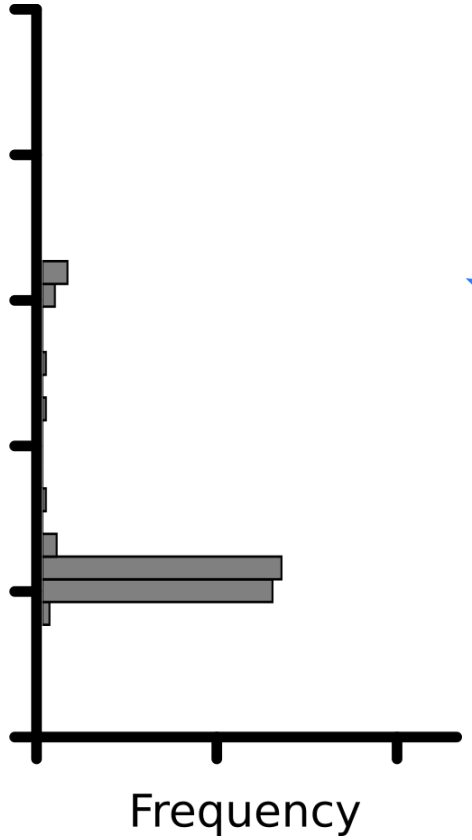



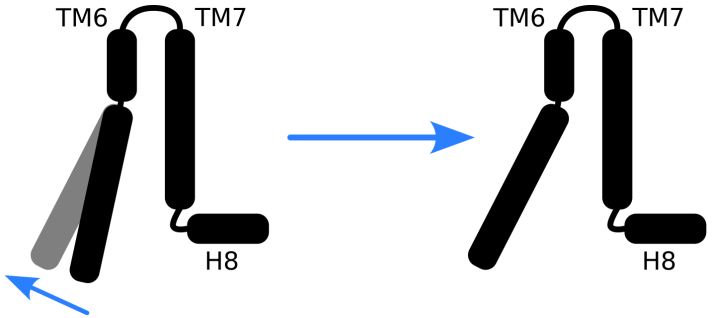



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# Sampling of Receptor Conformations



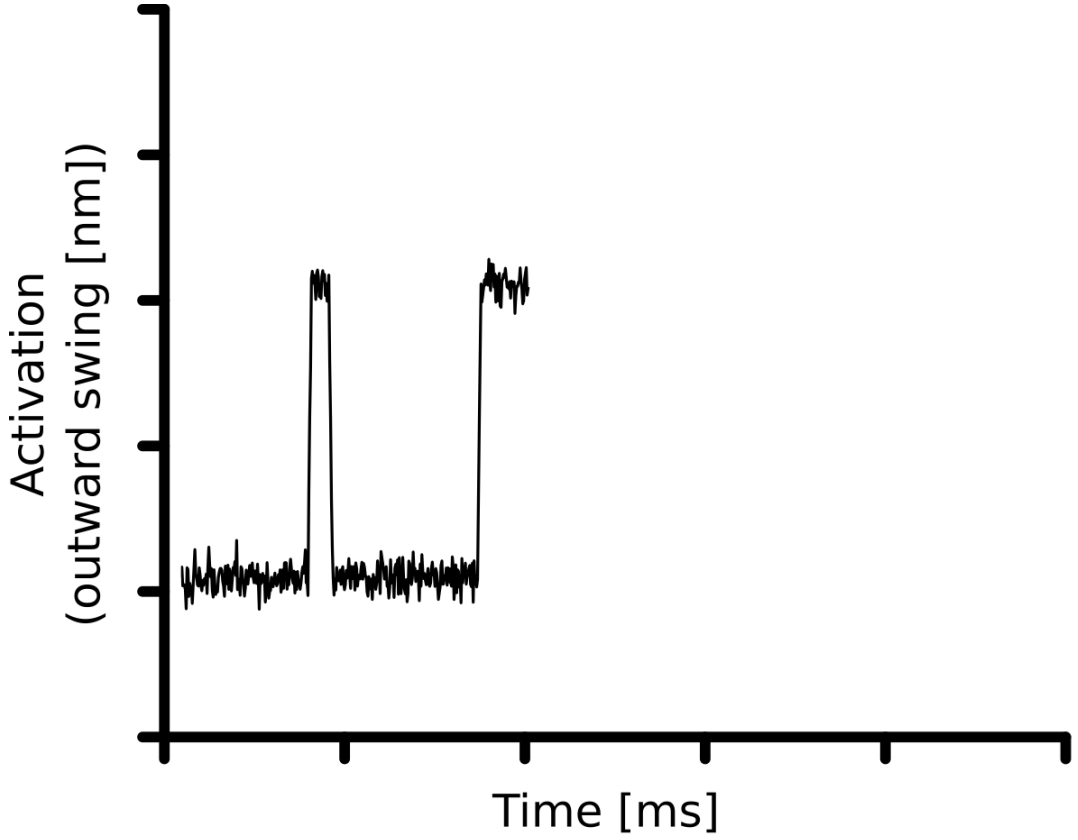



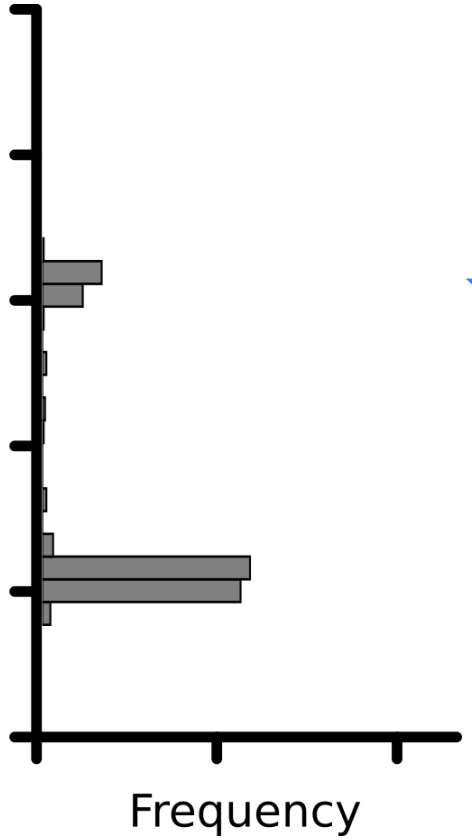



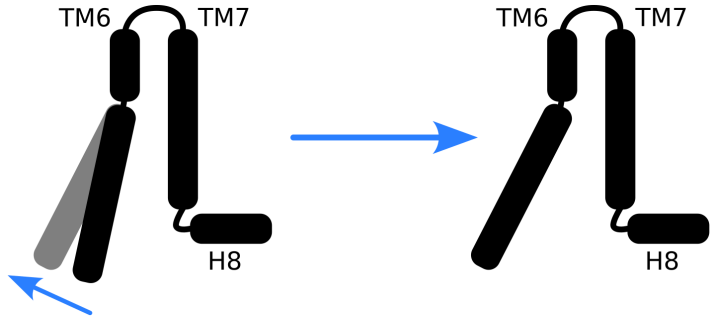



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# Sampling of Receptor Conformations



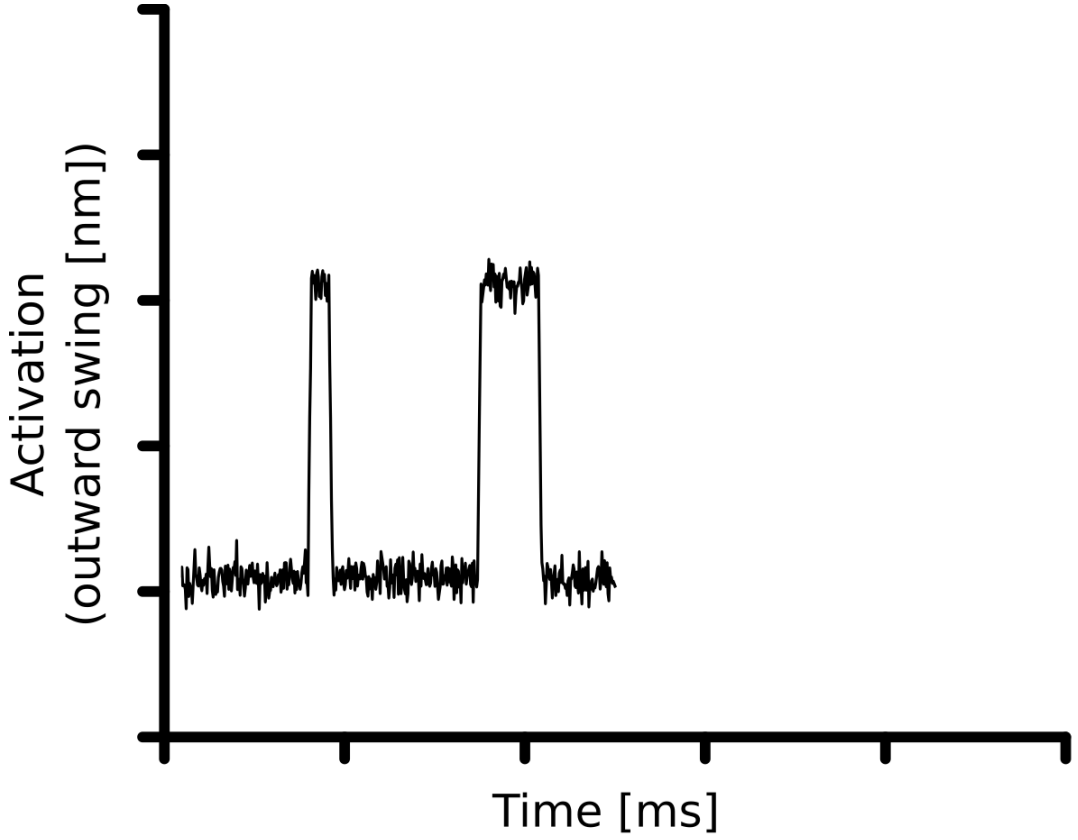



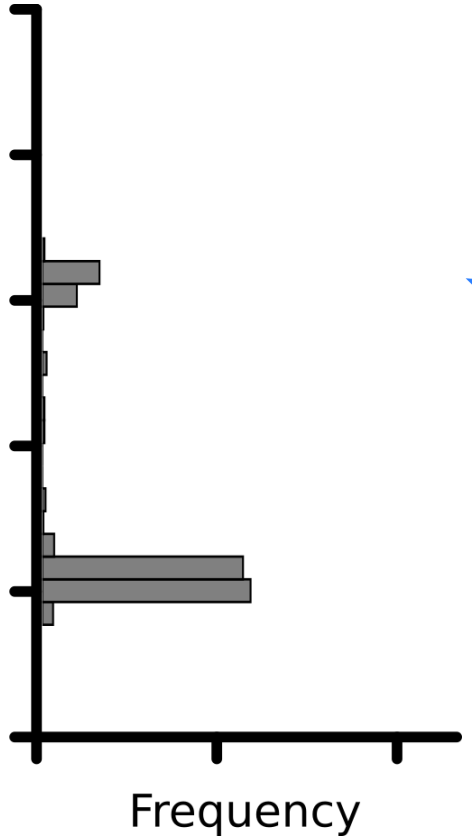



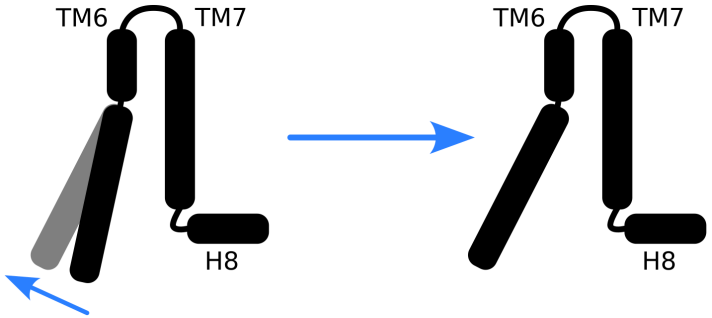



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# Sampling of Receptor Conformations



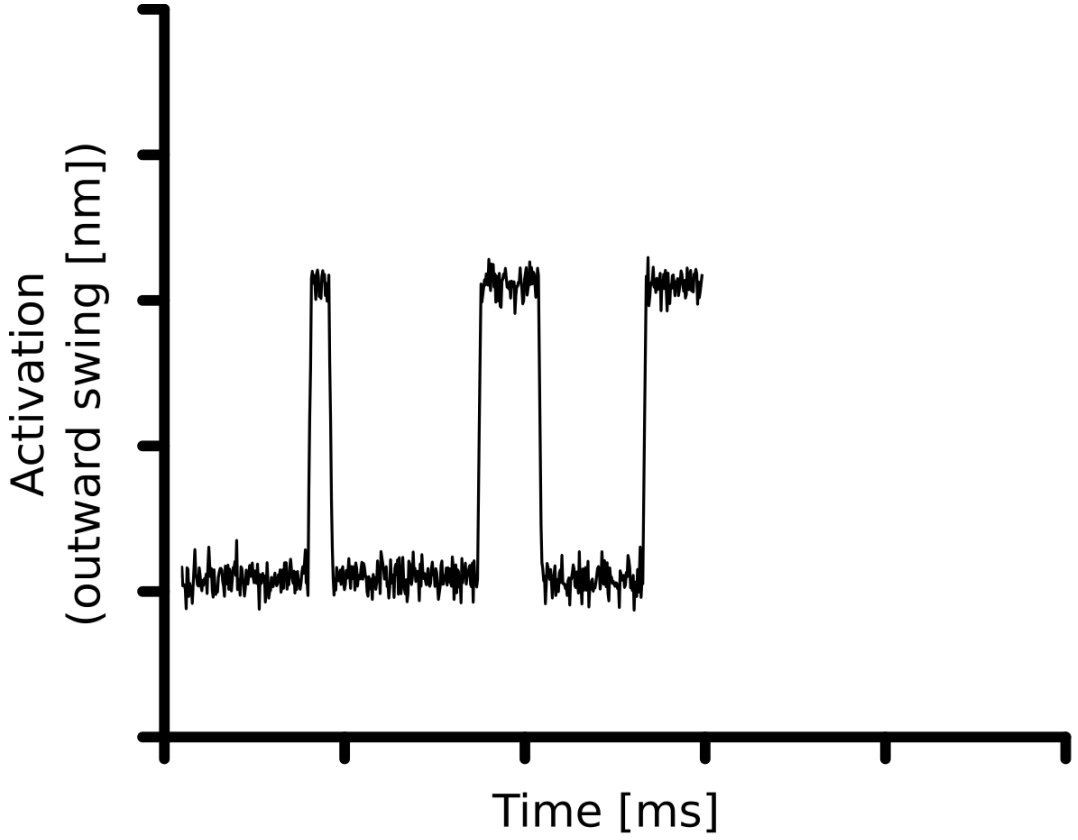



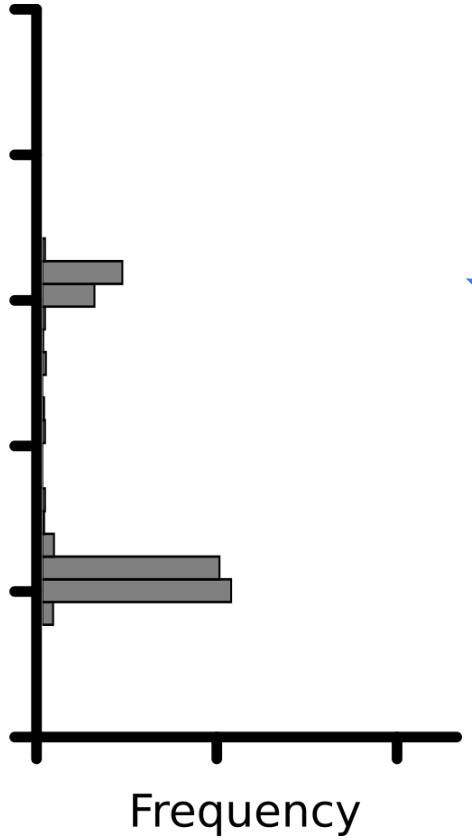



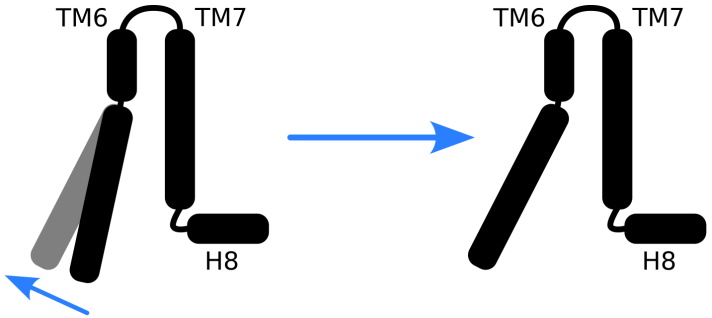



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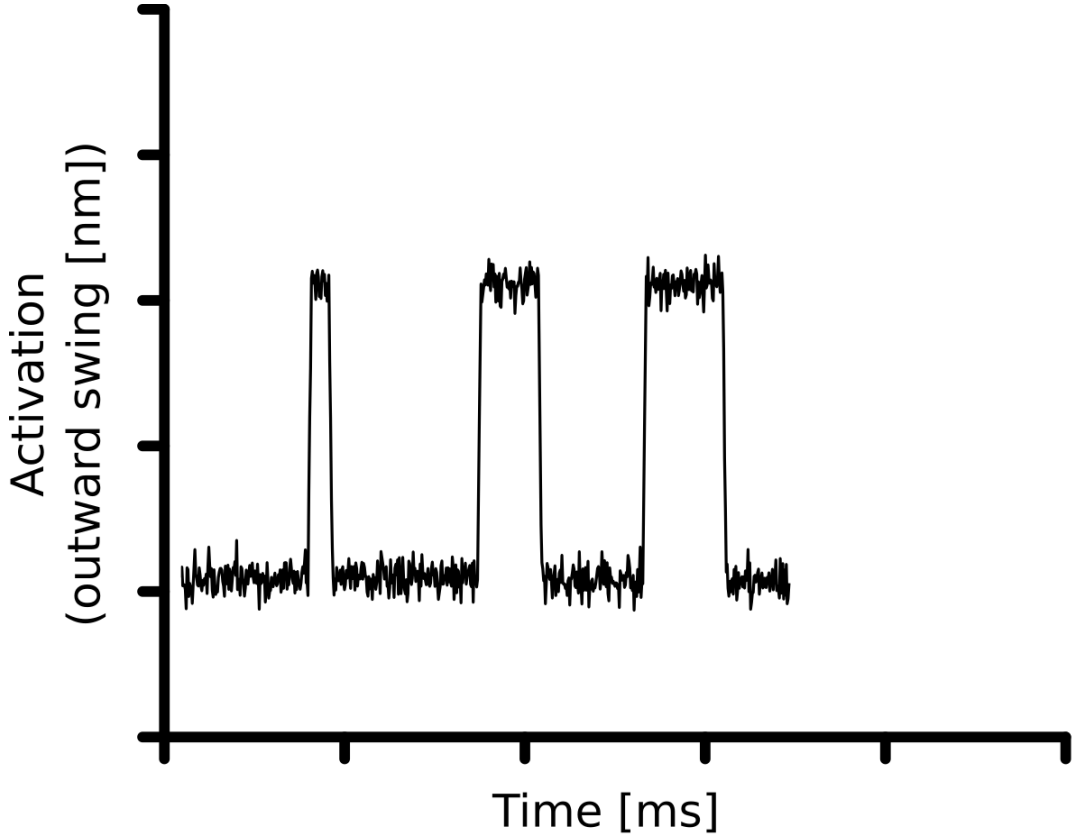



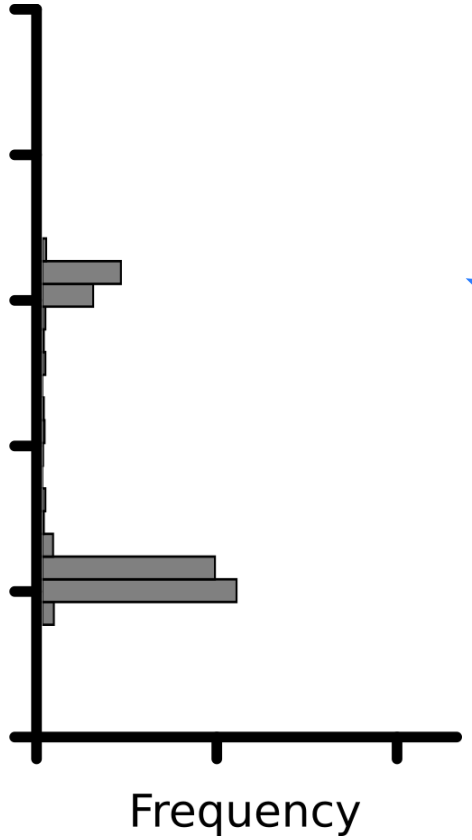



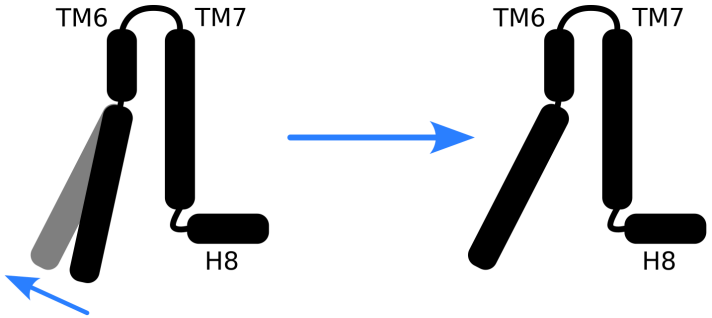



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# Sampling of Receptor Conformations



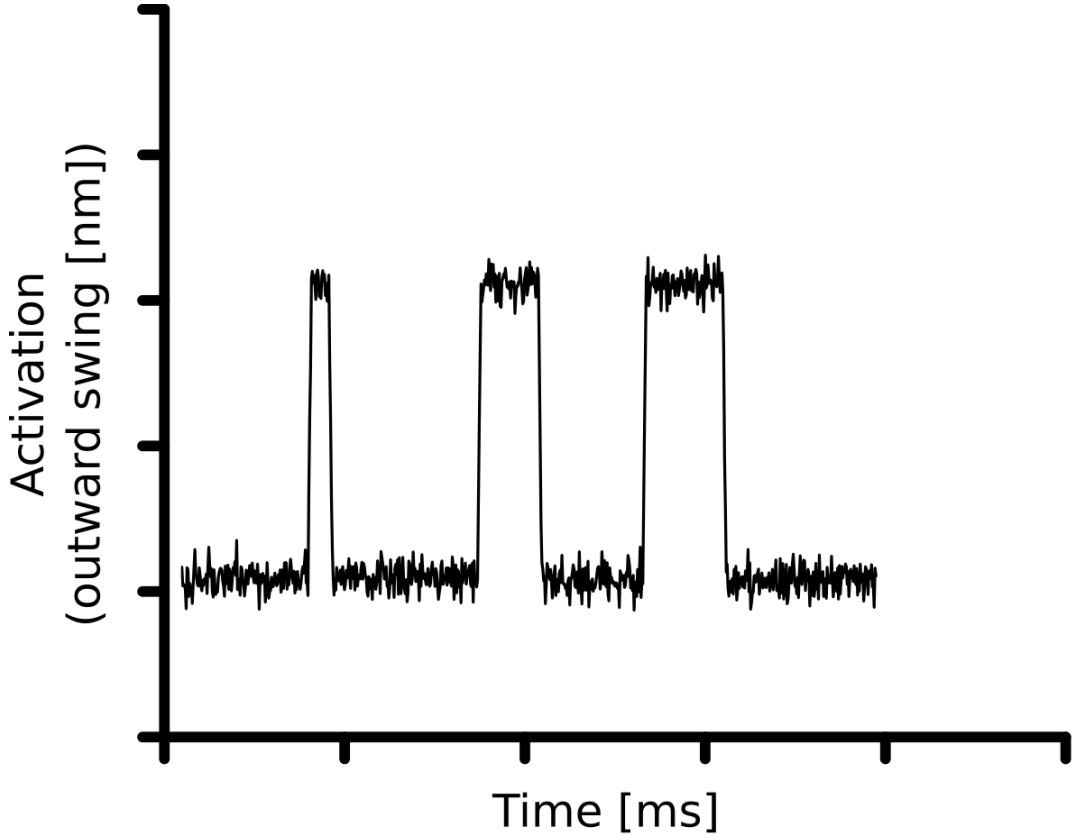



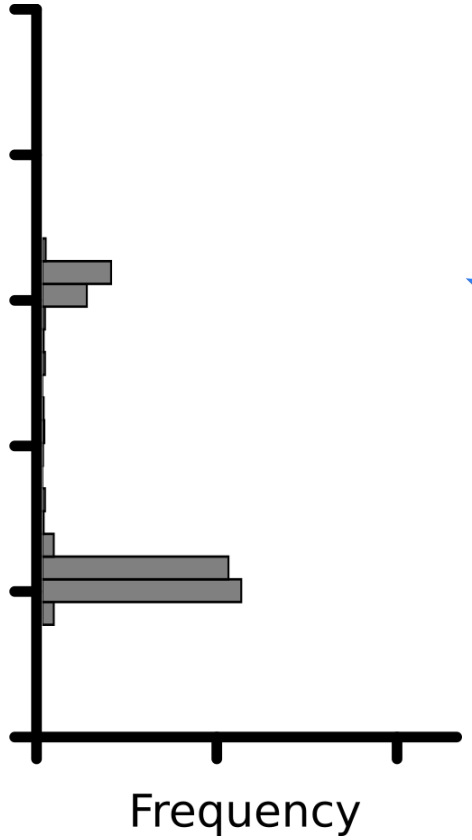



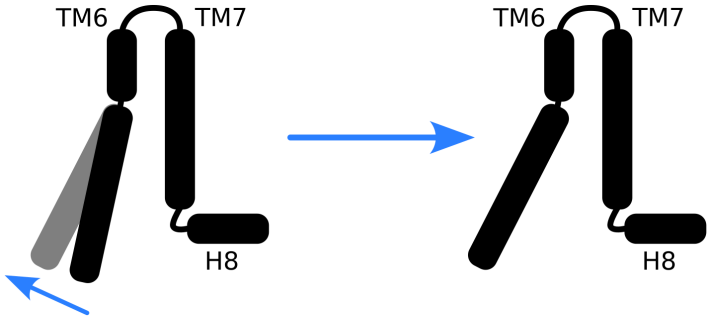



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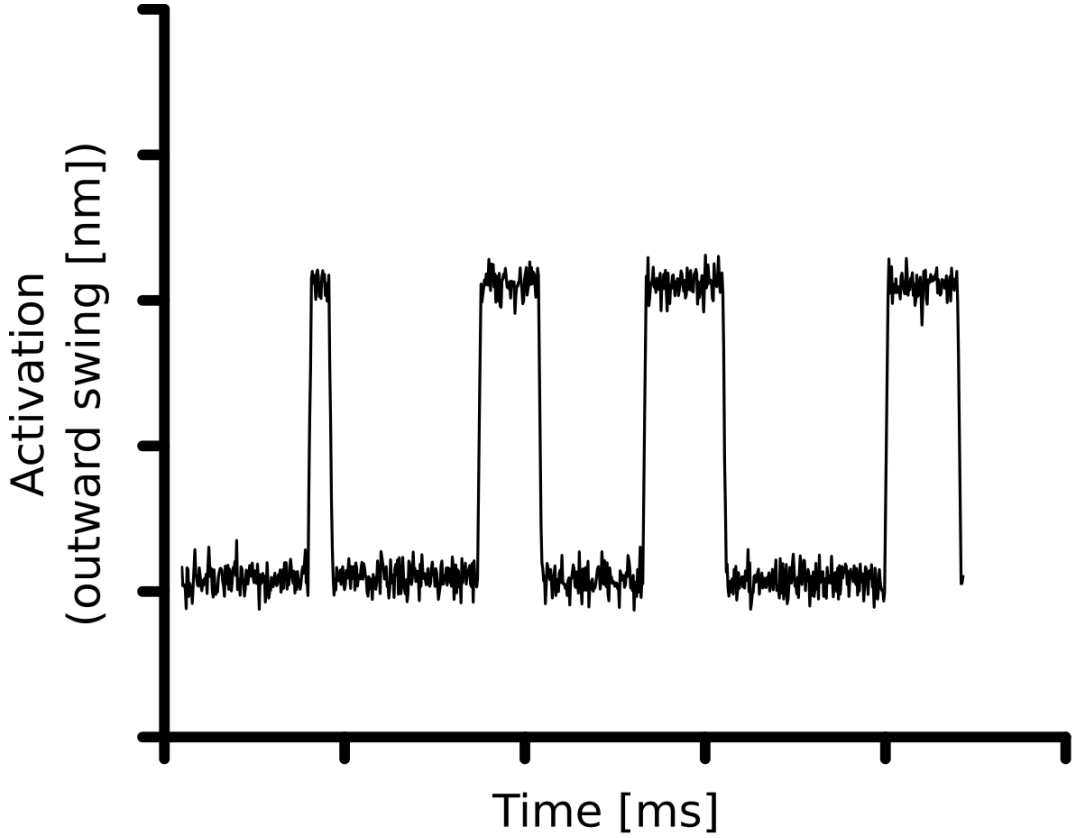



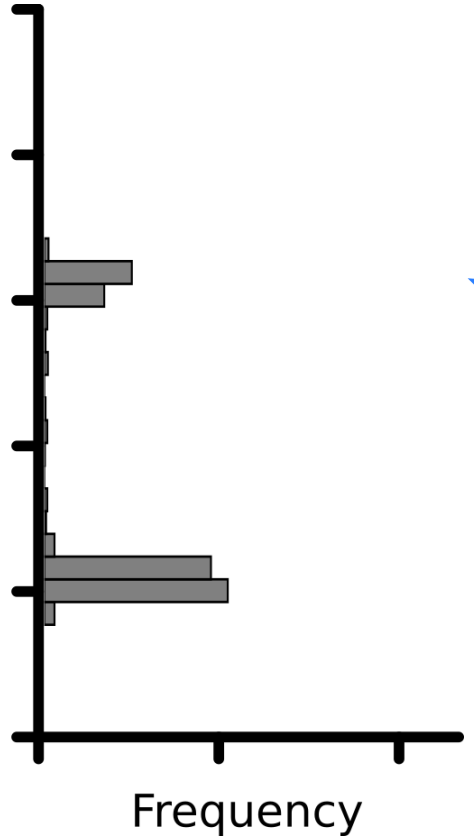



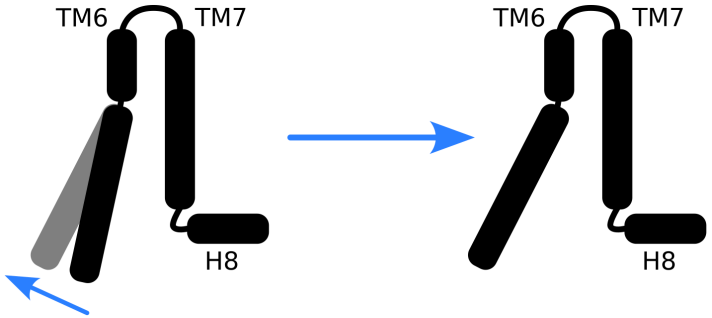



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# Sampling of Receptor Conformations



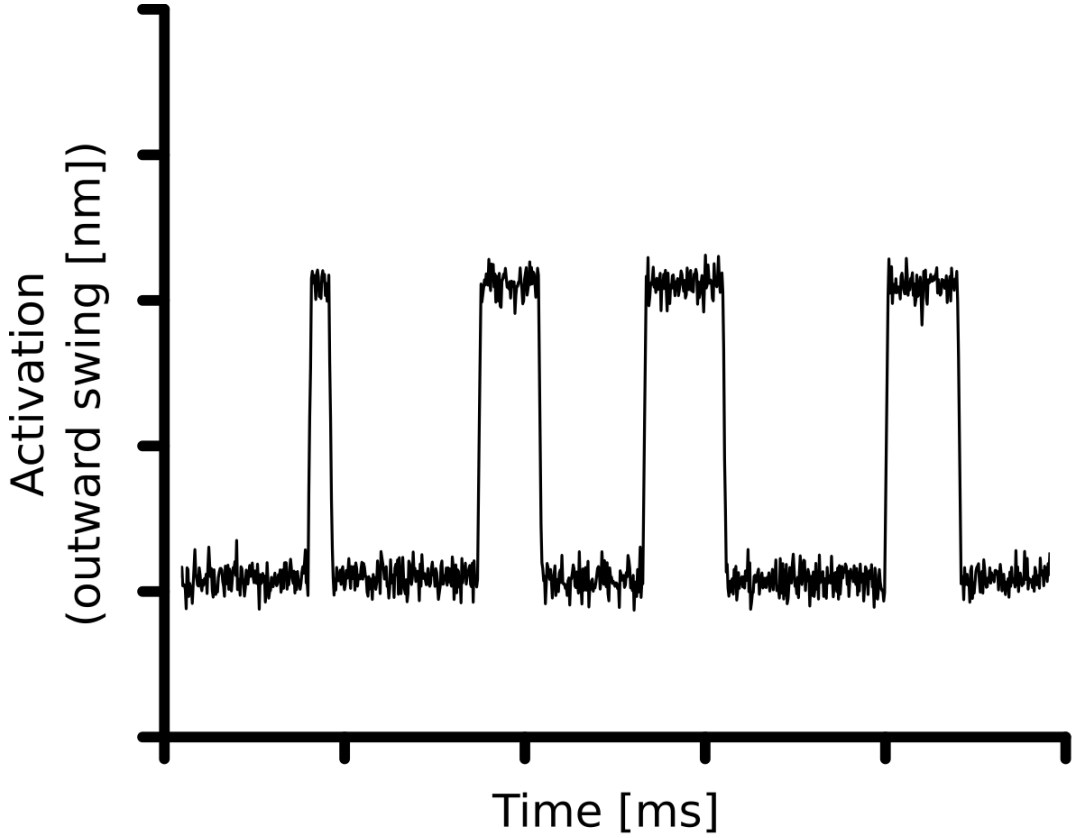



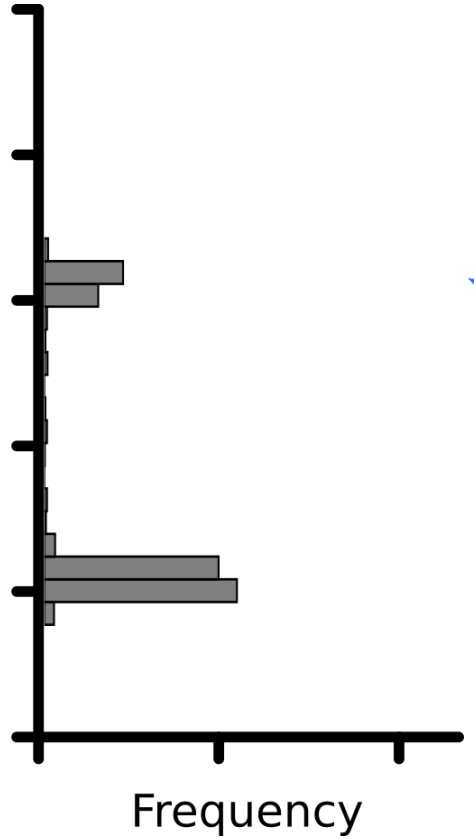



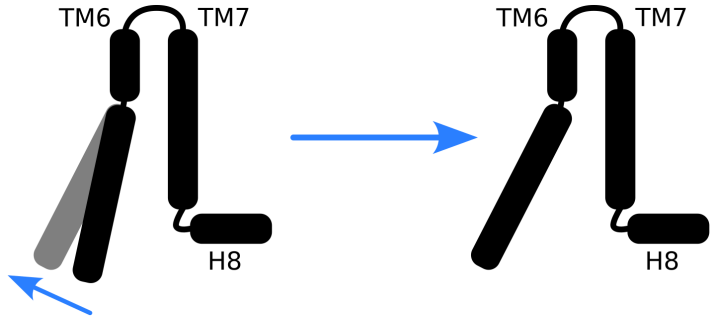



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# Sampling of Receptor Conformations



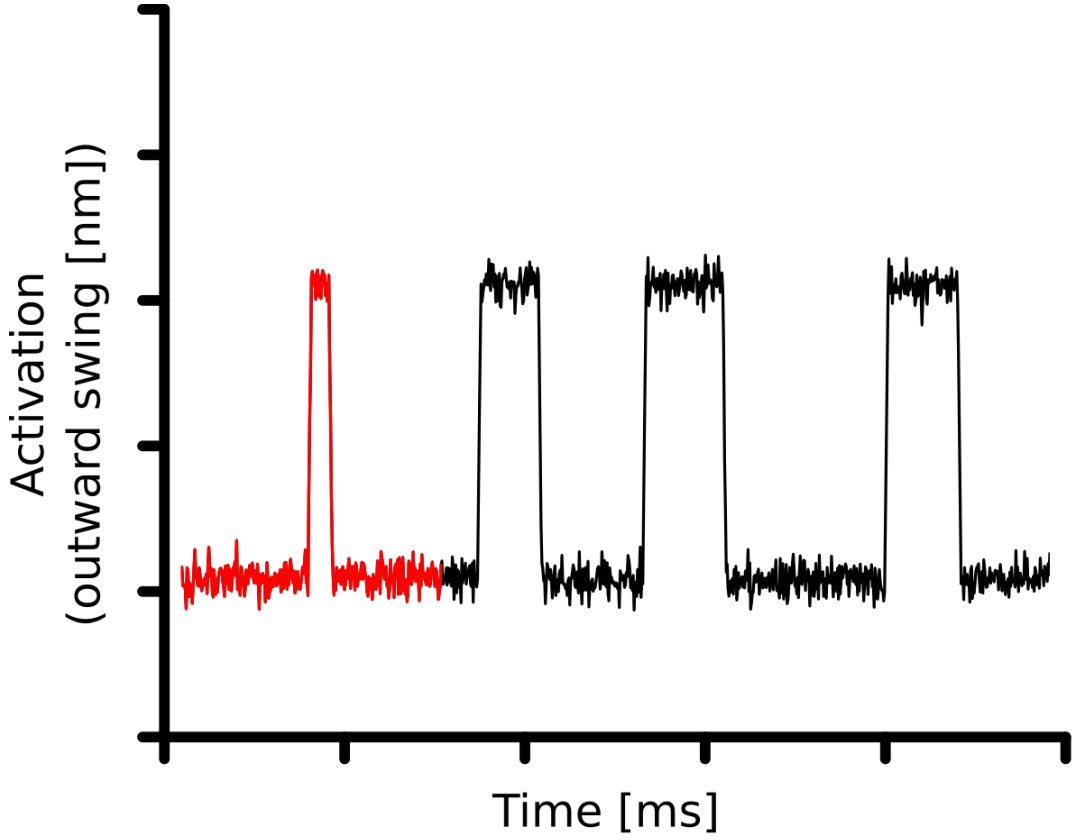



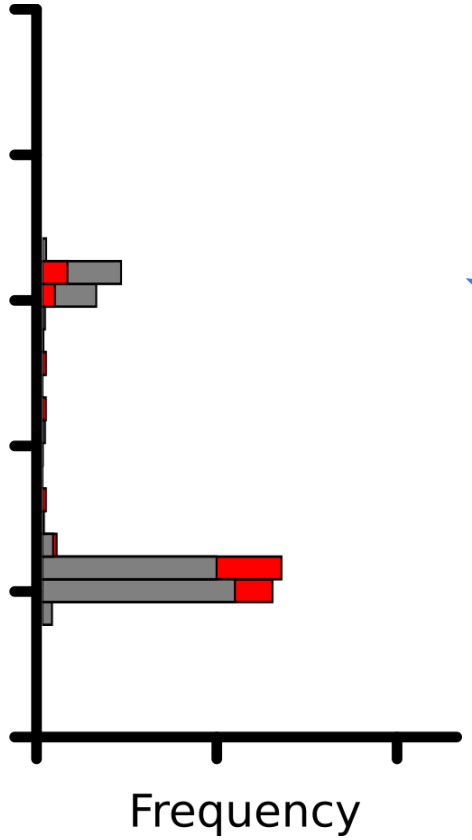



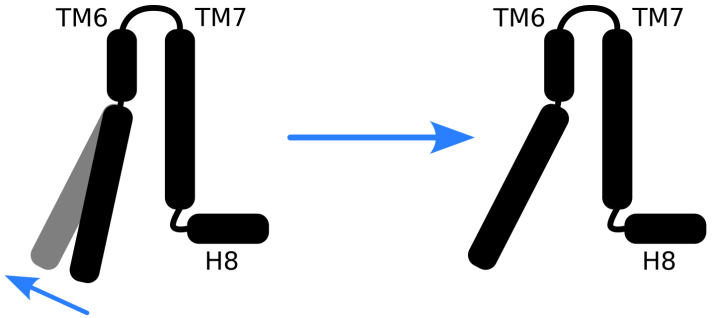



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# Sampling of Receptor Conformations



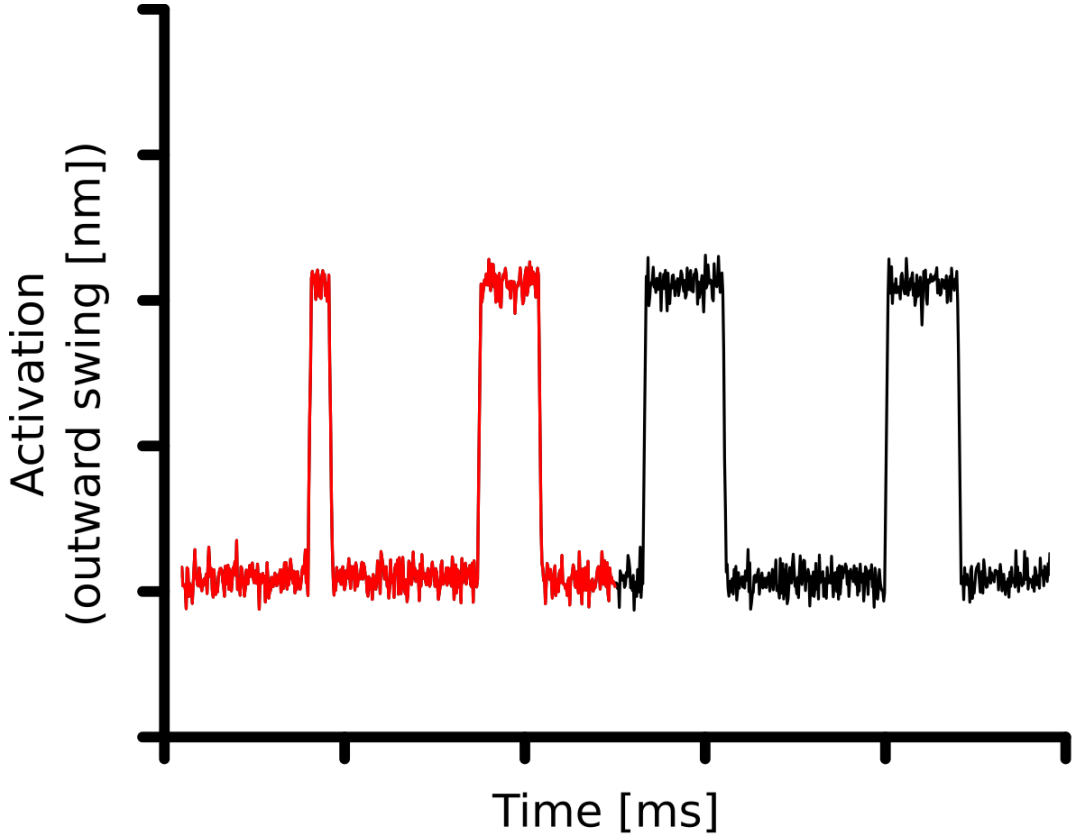



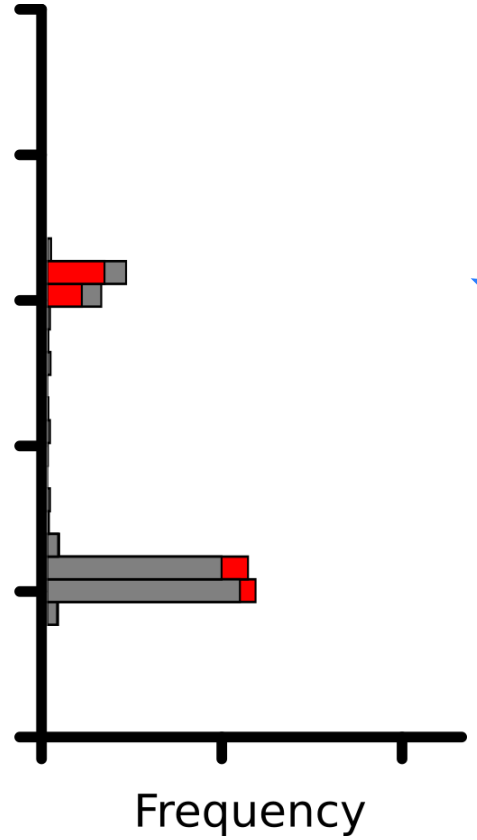



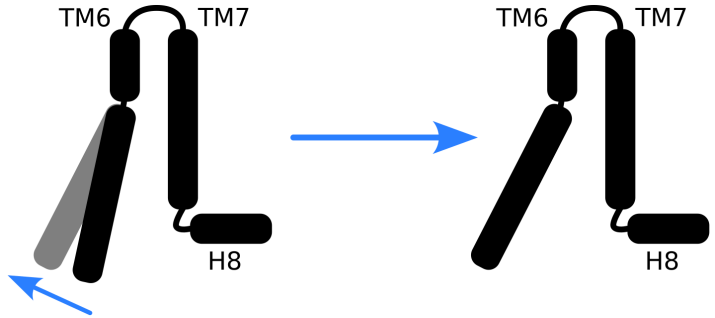



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# Sampling of Receptor Conformations



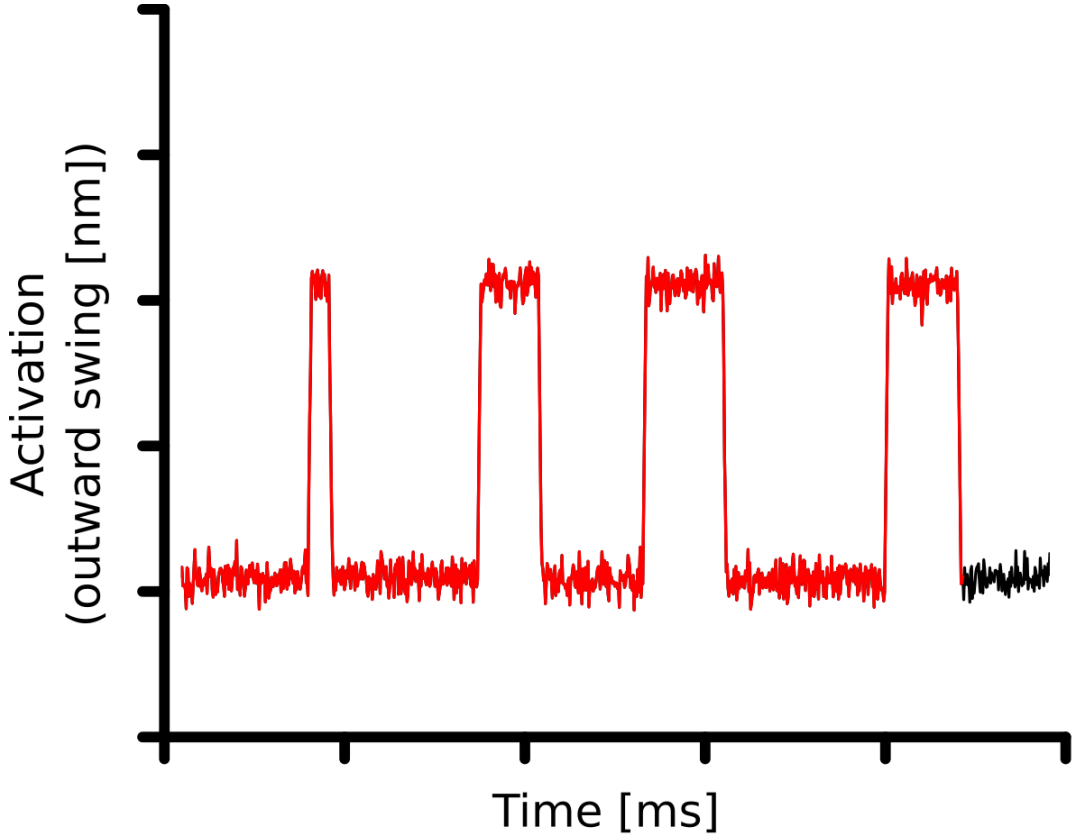



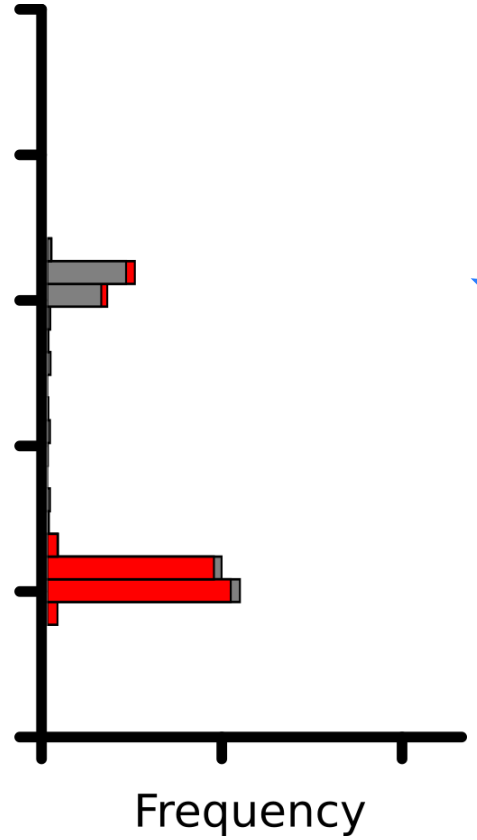



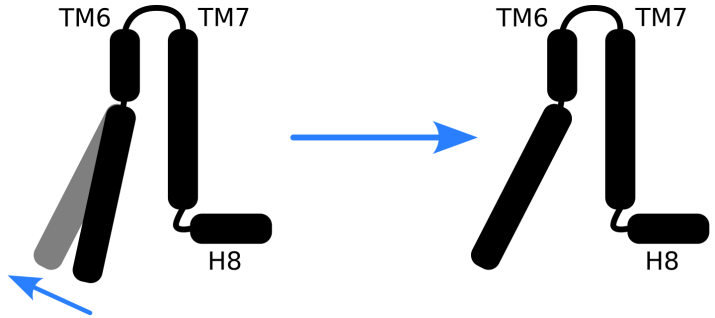



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# Sampling of Receptor Conformations



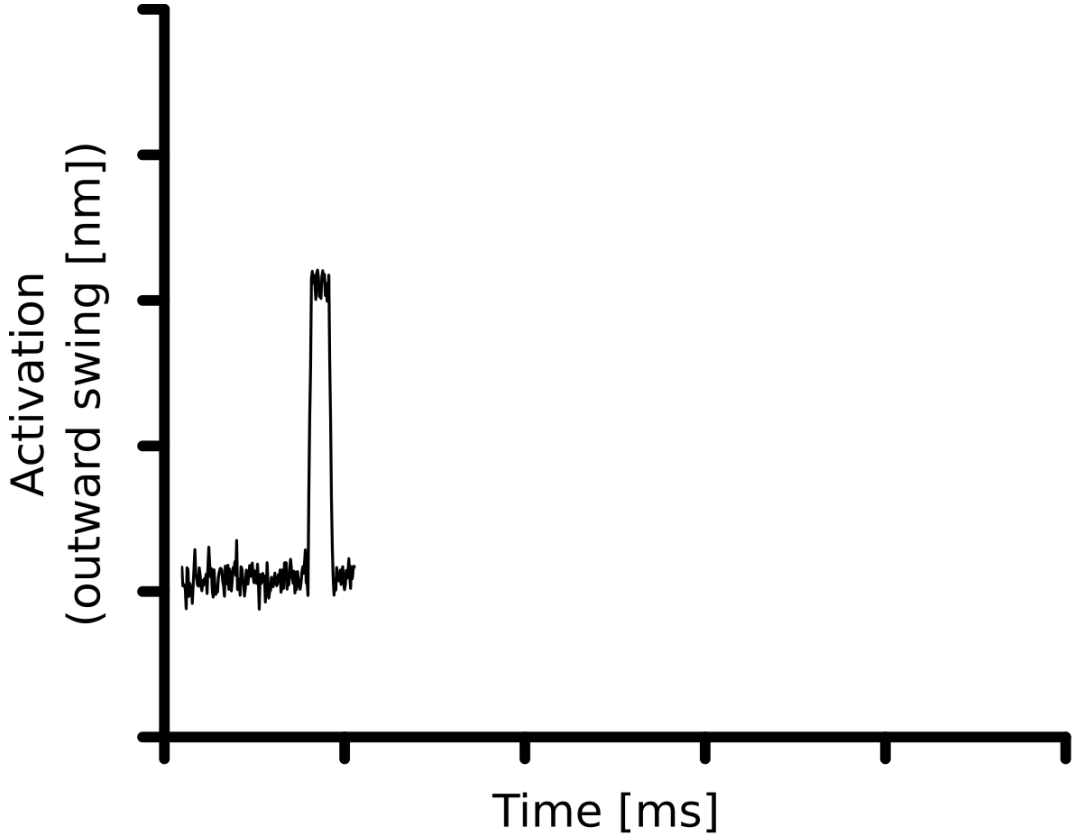


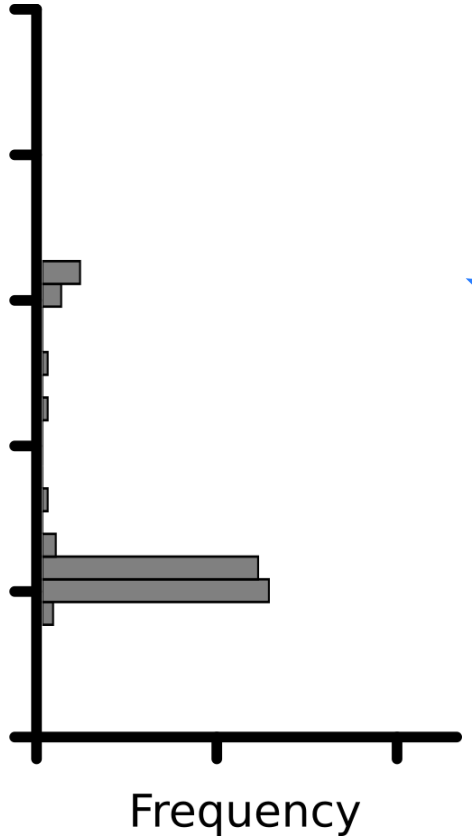

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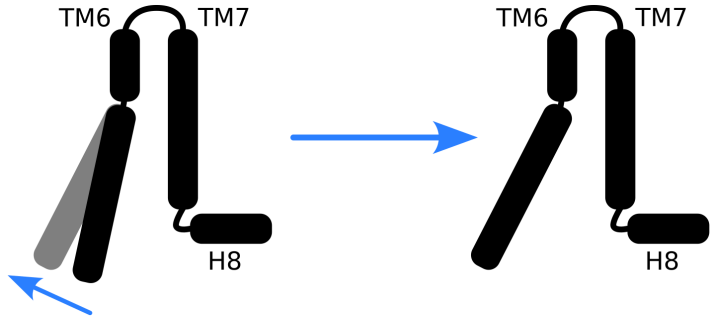
# Convergence Tests

- Increase simulation length
  - Monitor if time-averaged quantities change
- Block averaging
  - Divide a trajectory into blocks of equal length and calculate averages for each block
  - Stratification / strata
- Multiple starting conditions
  - Simulations should yield similar averages
- Check for symmetries

# Sampling of Receptor Conformations



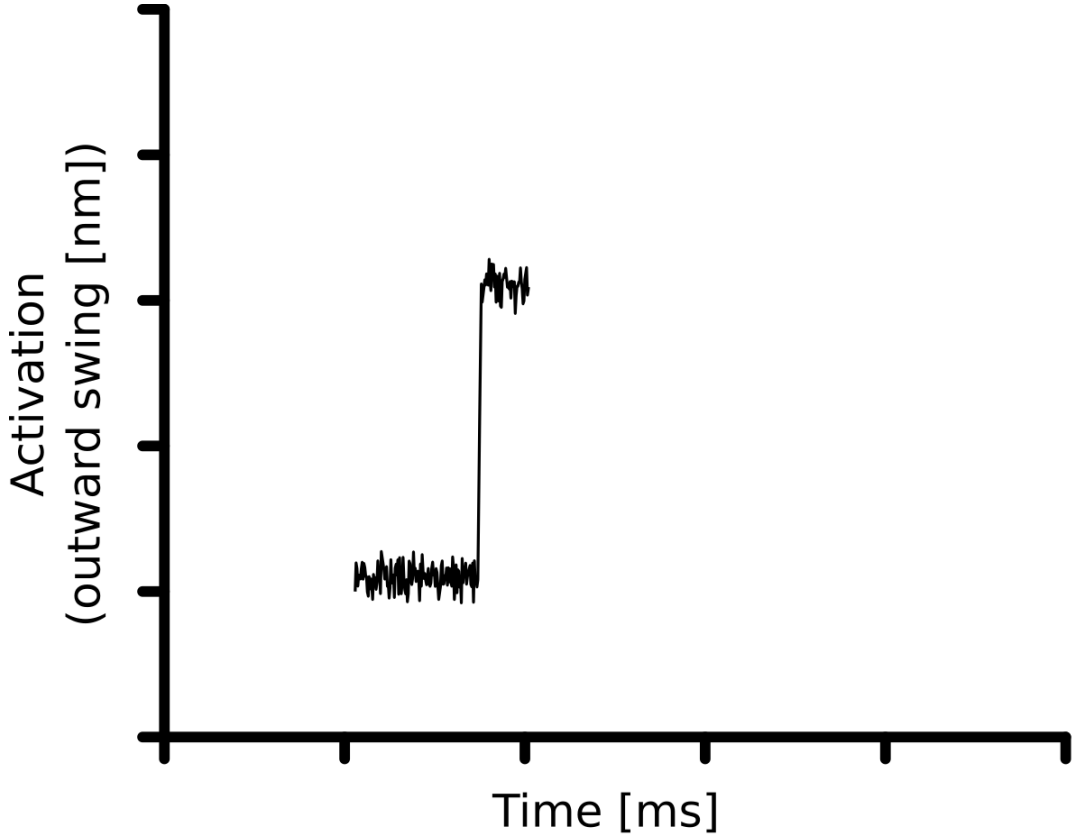



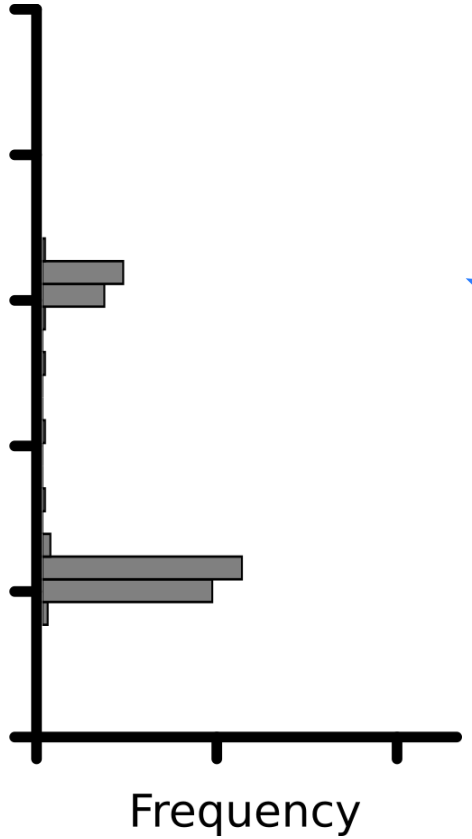



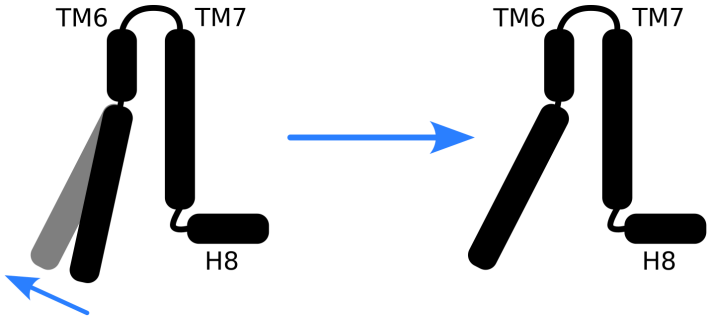



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# Sampling of Receptor Conformations



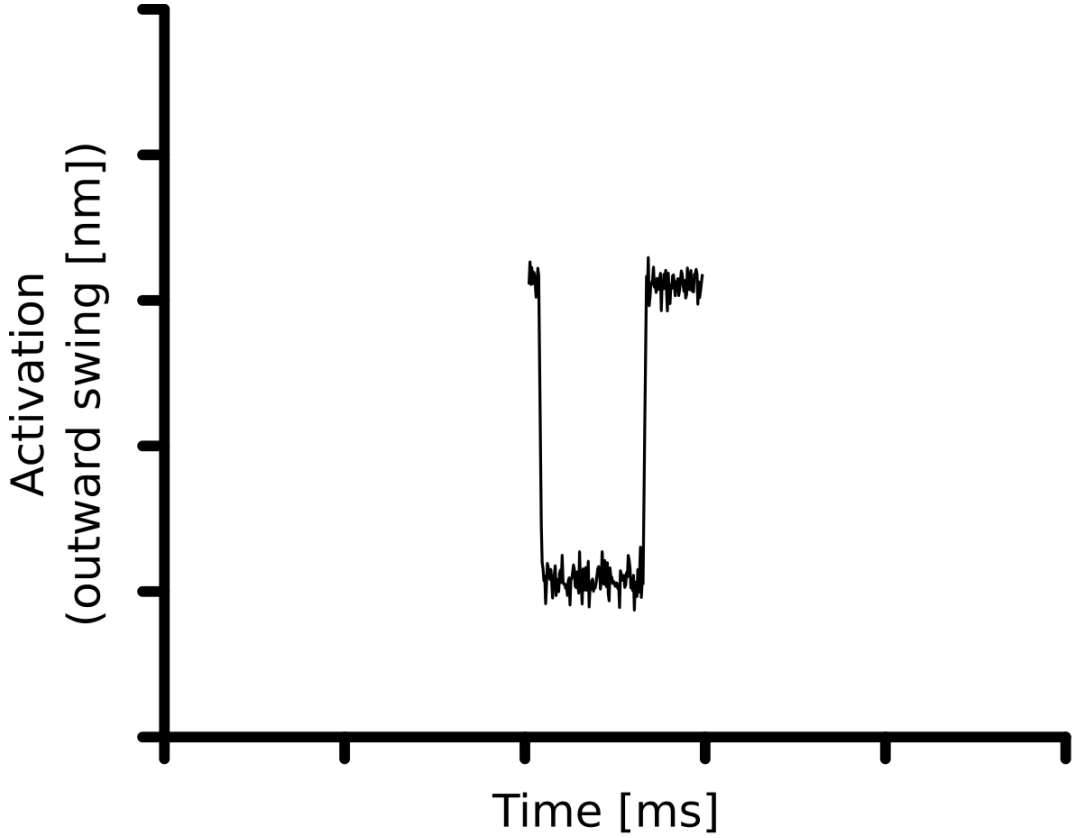



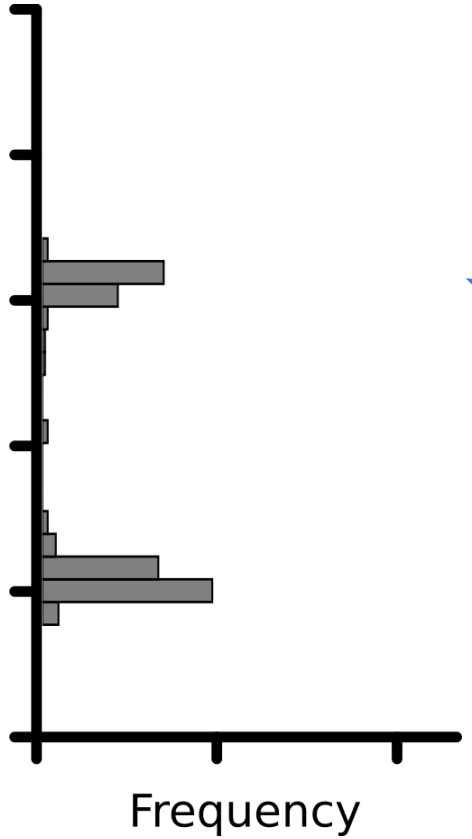



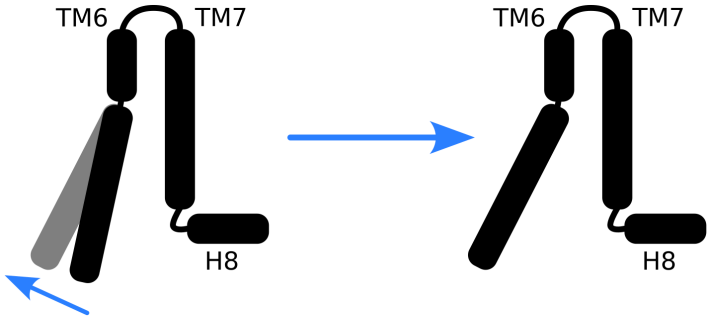



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# Sampling of Receptor Conformations



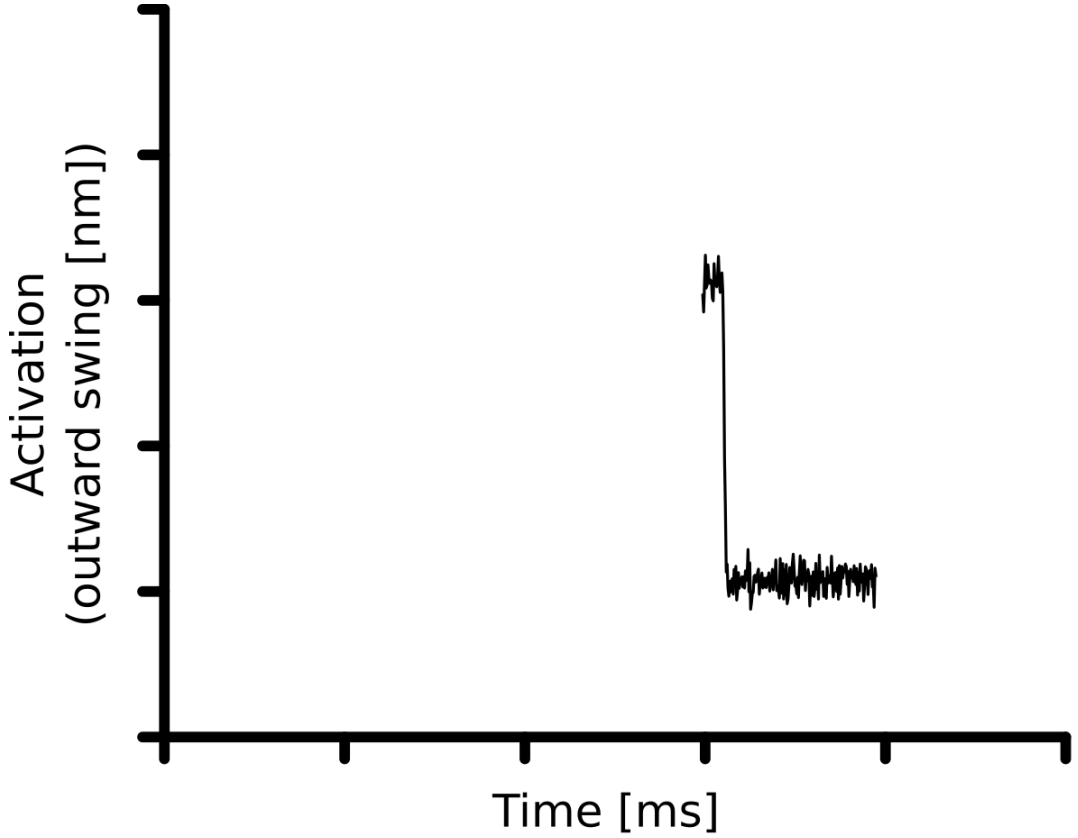



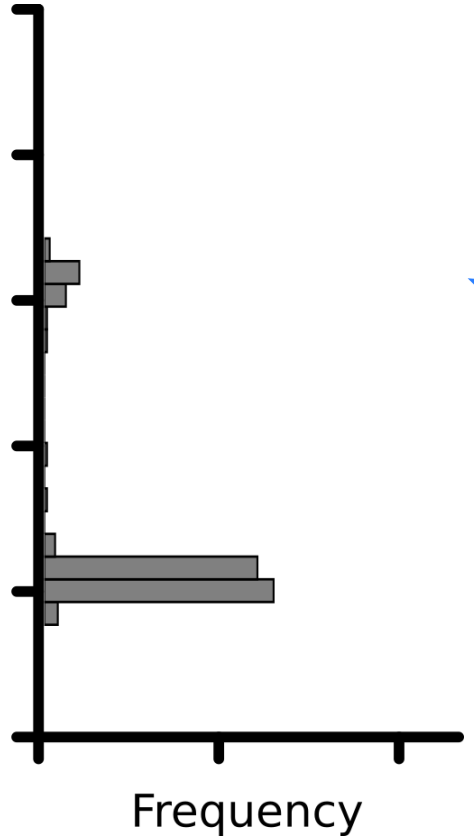



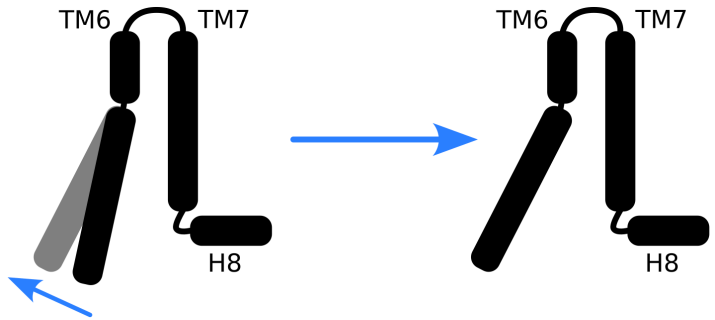



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# Sampling of Receptor Conformations



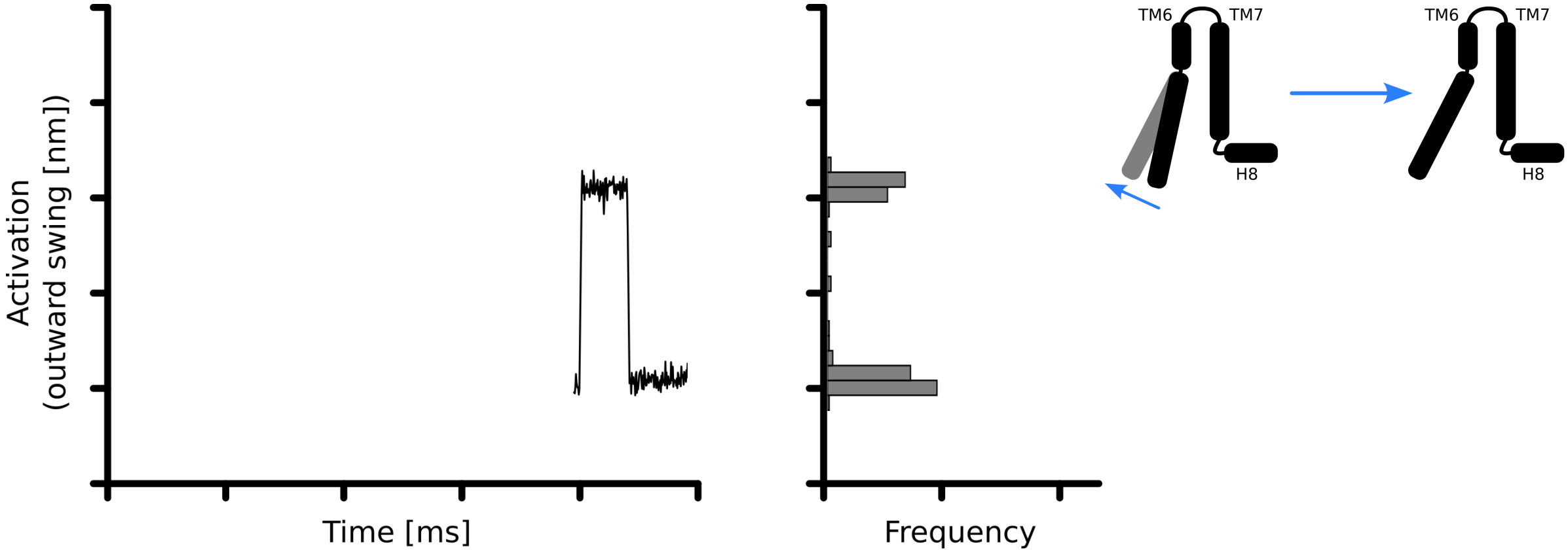






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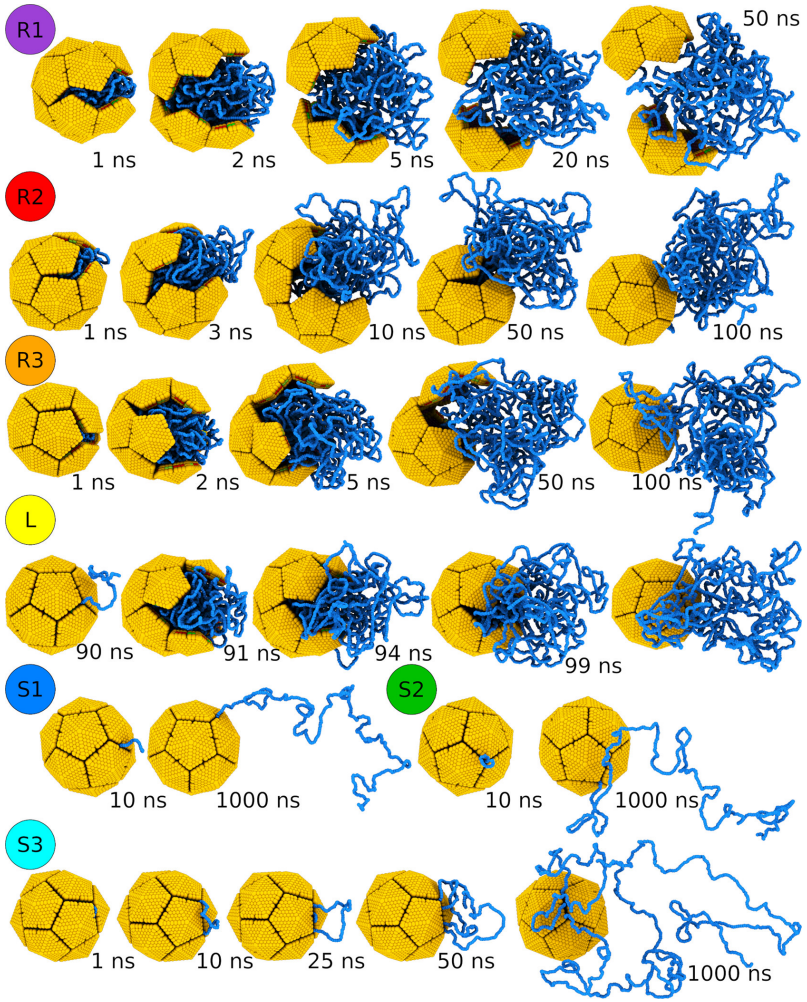
# Sampling of Receptor Conformations






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# DOI: 10.1021/acsnano.1c04814



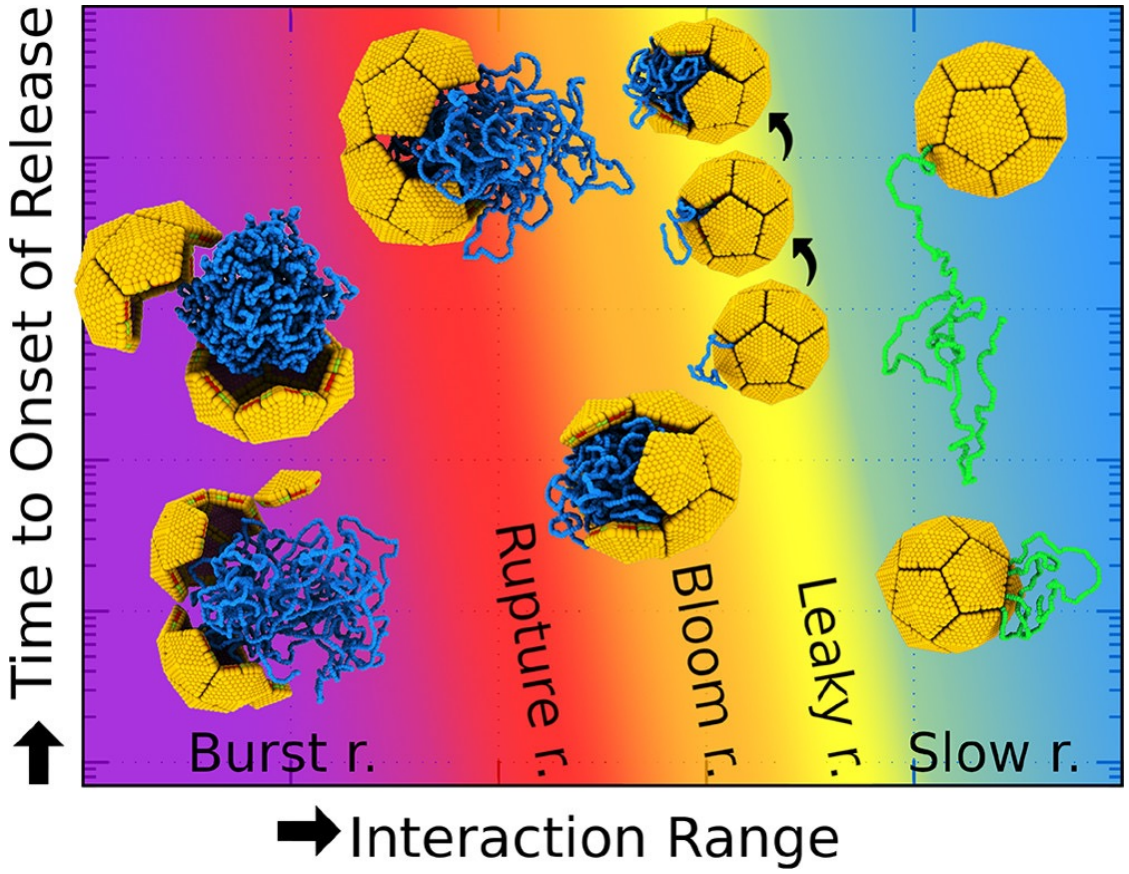






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# DOI: 10.1021/acsnano.1c04814



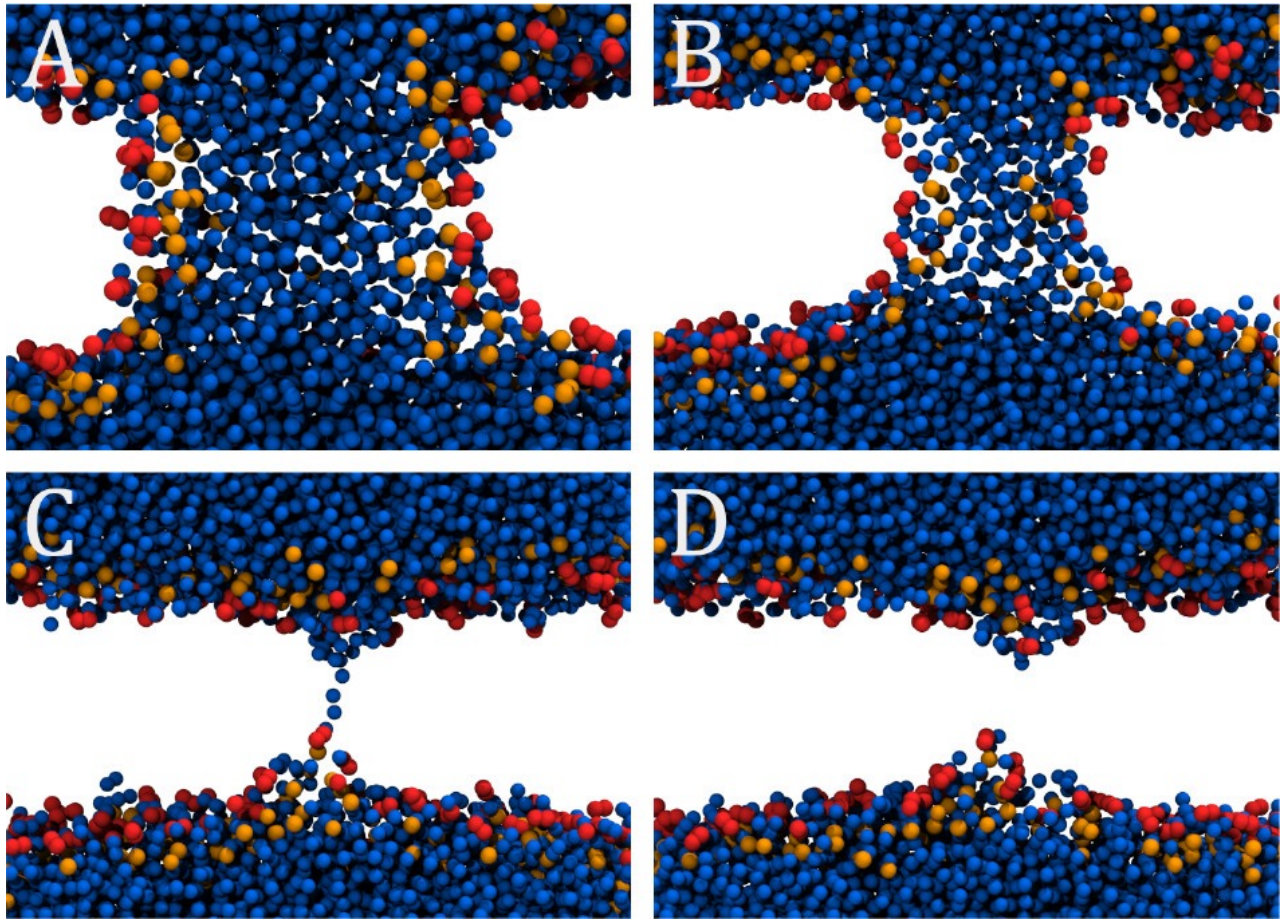


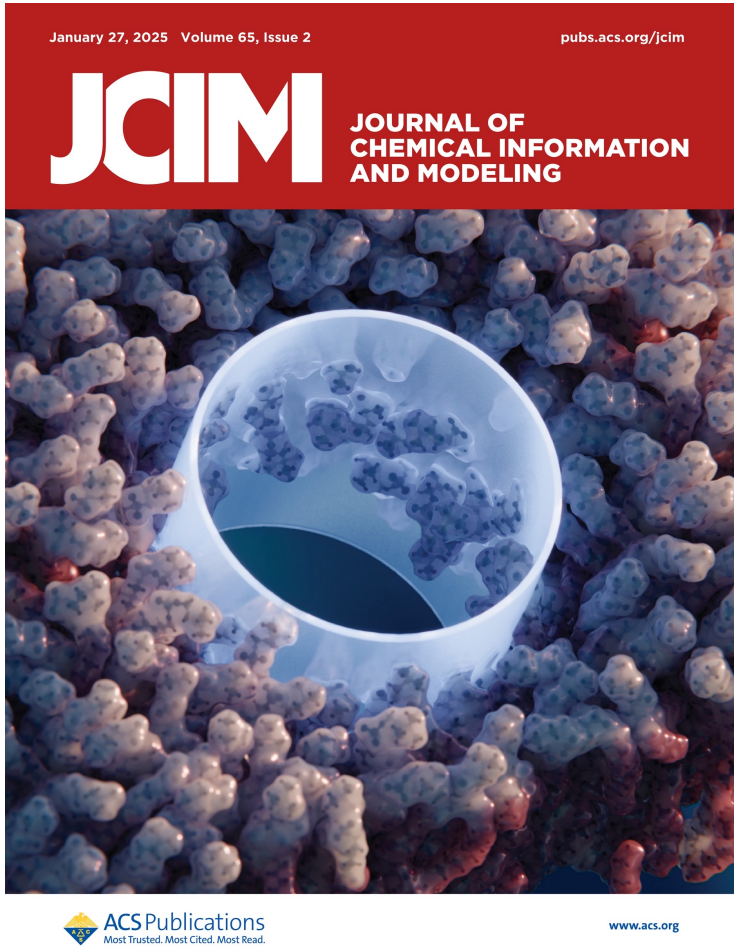




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# DOI: 10.1021/acs.jcim.4c01960

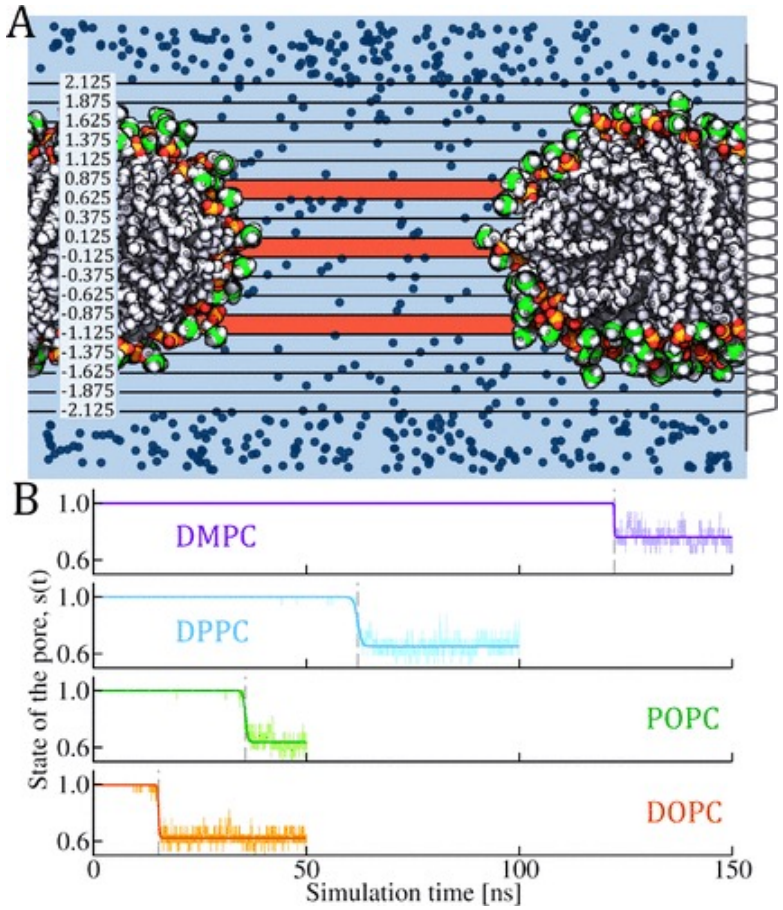


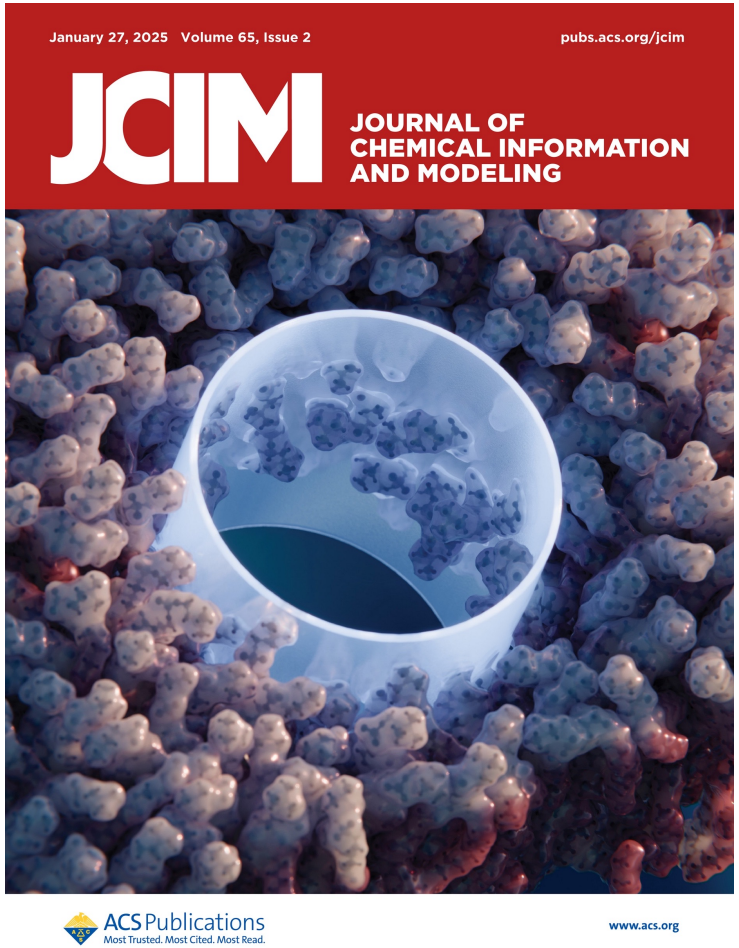






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# DOI: 10.1021/acs.jcim.4c01960



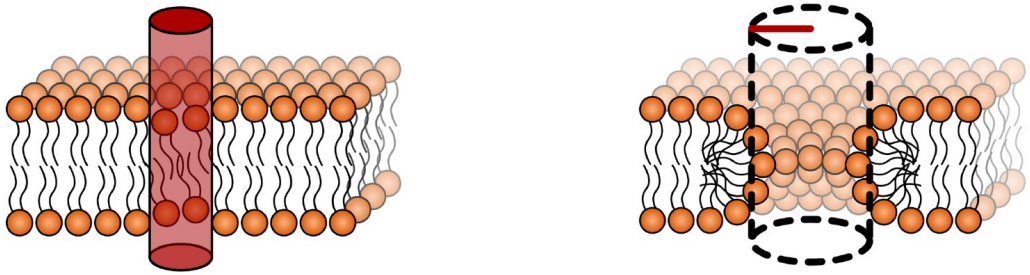






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# DOI: 10.1021/acs.jcim.4c01960

**Joint Collective Variable for "Full-Path" Method**



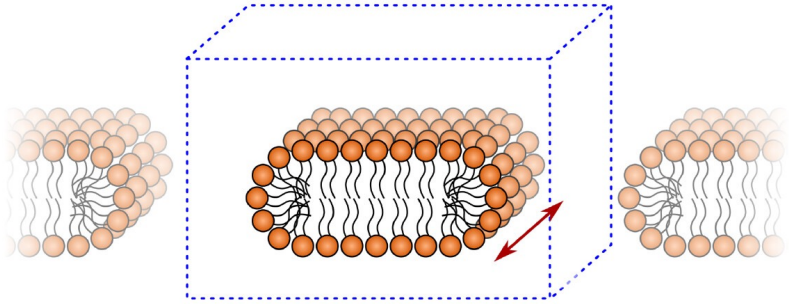
Part I: Membrane Defect

$CV_{cyl}$ – number of lipid-tail heavy atoms within a cylindrical volume

Part II: Pore Expansion

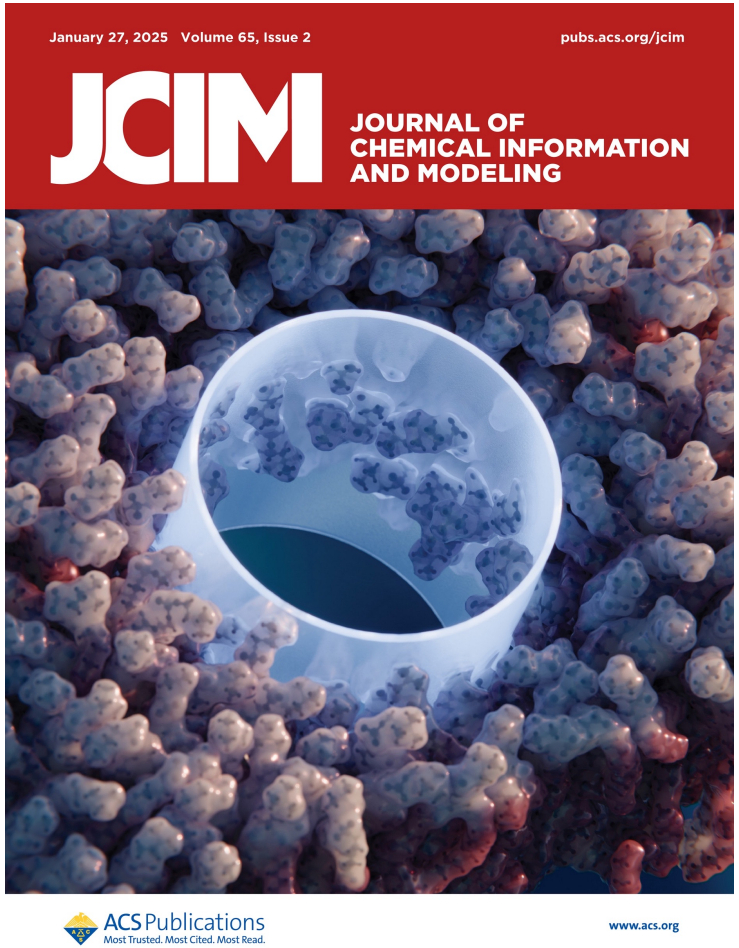
$CV_{radius}$ – smallest pore radius

**Collective Variable for "Rapid" Method**



Pore Expansion

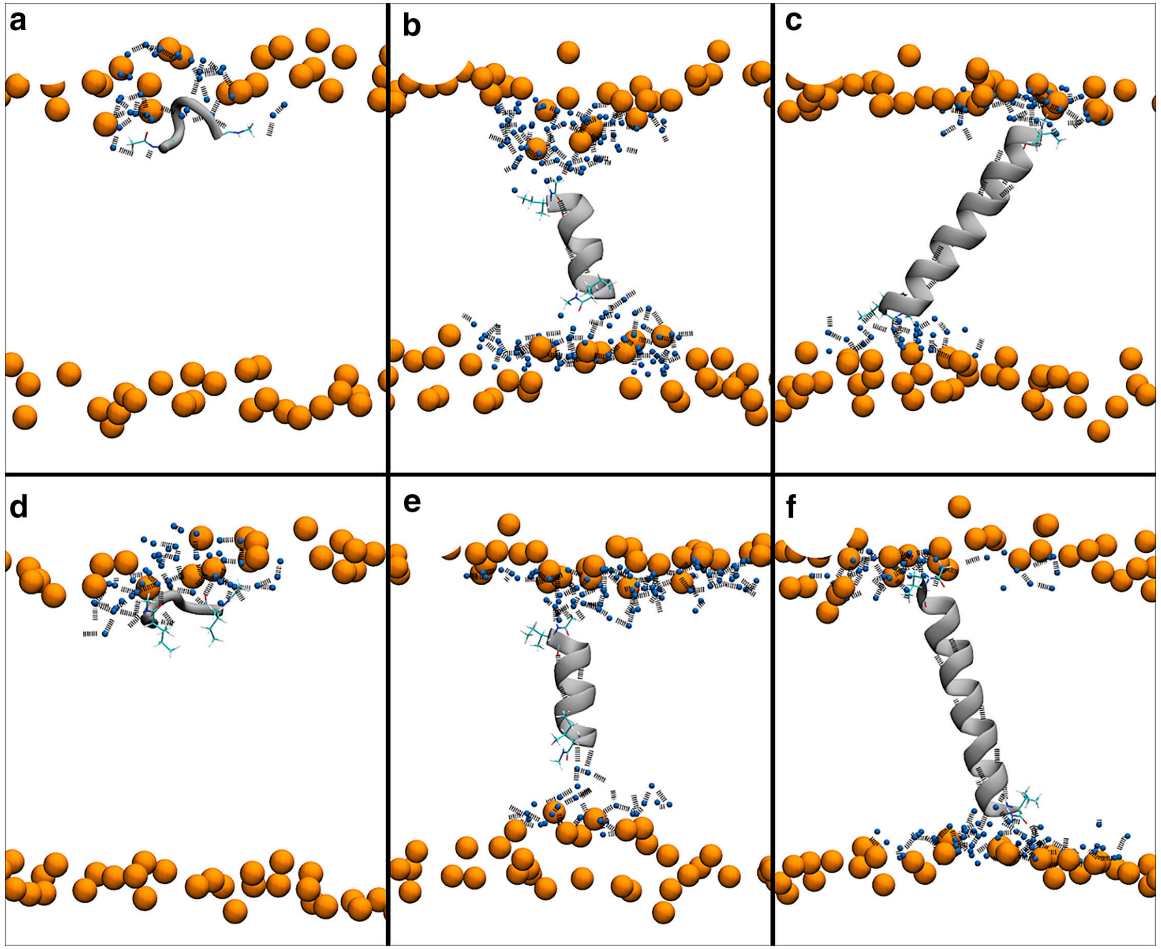
$CV_{box}$ – box size along the membrane edge






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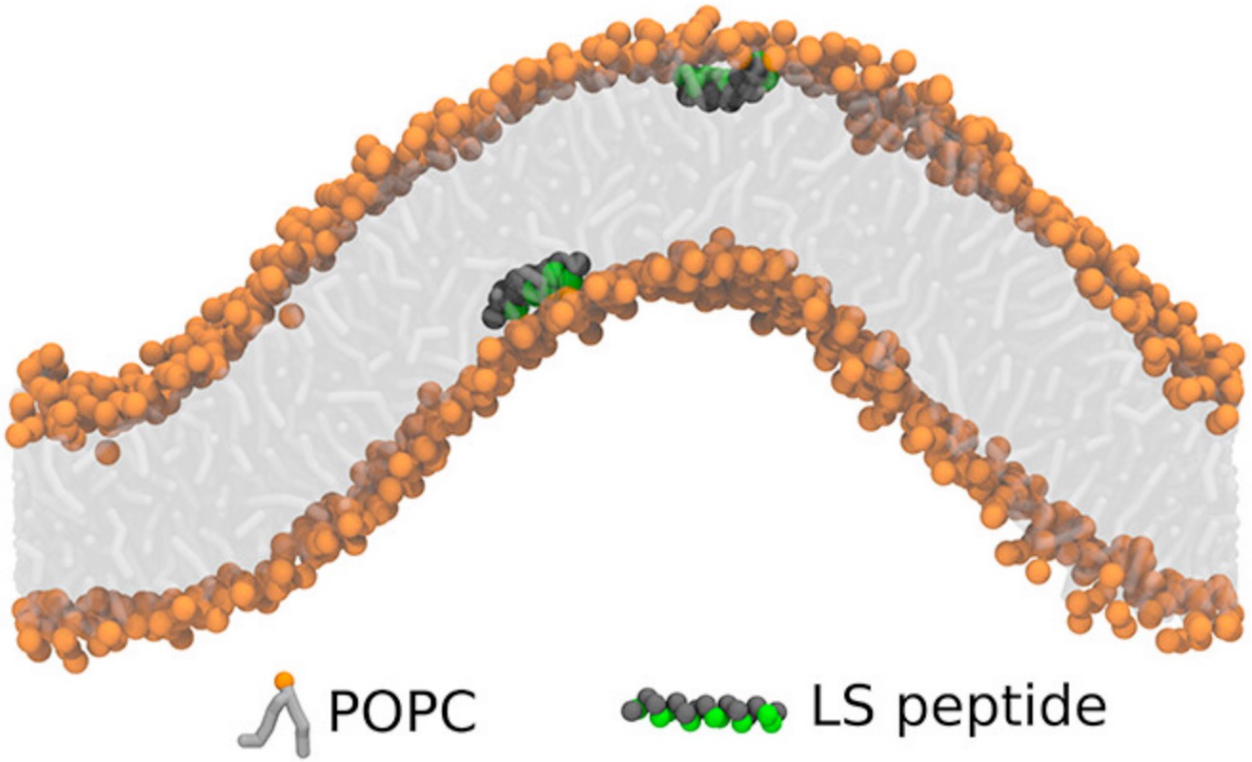
# DOI: 10.1016/j.bpj.2018.08.012

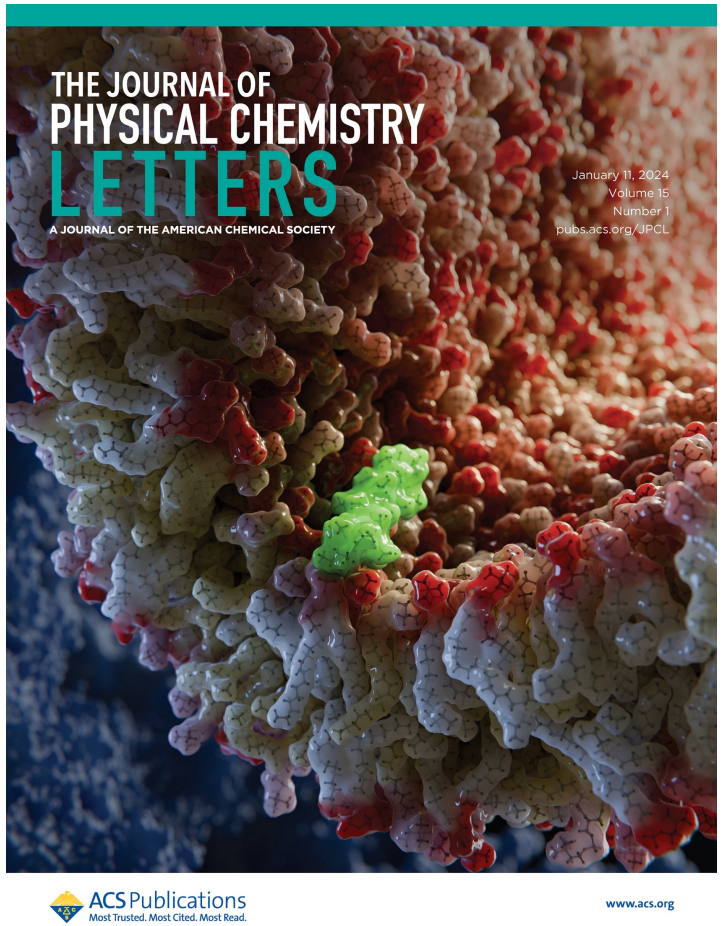





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# DOI: 10.1021/acs.jpclett.3c02785

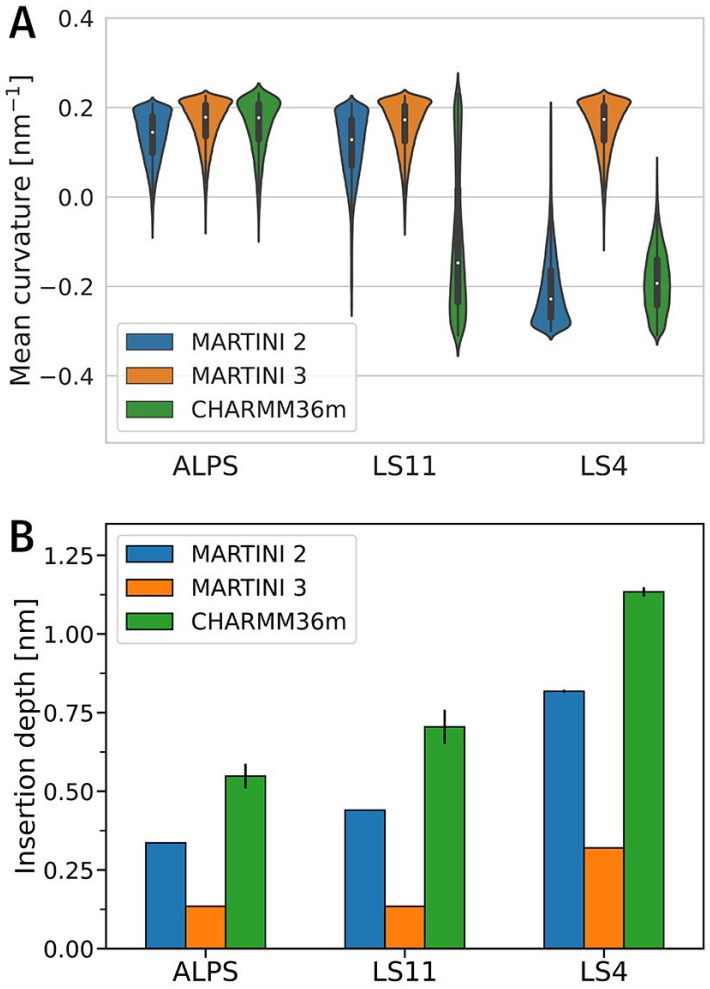


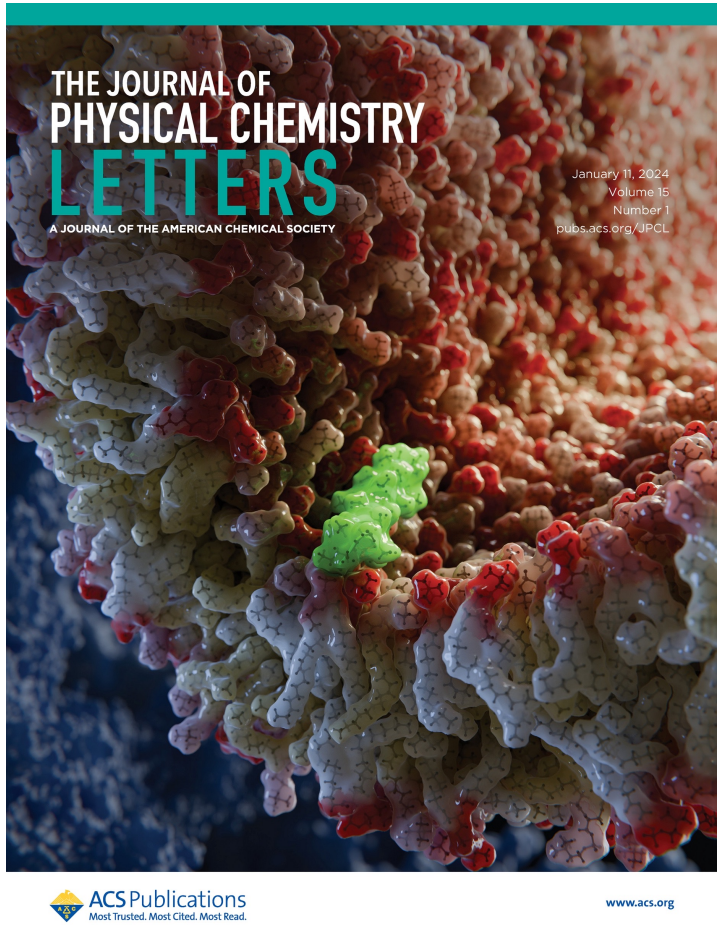






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# DOI: 10.1021/acs.jpclett.3c02785

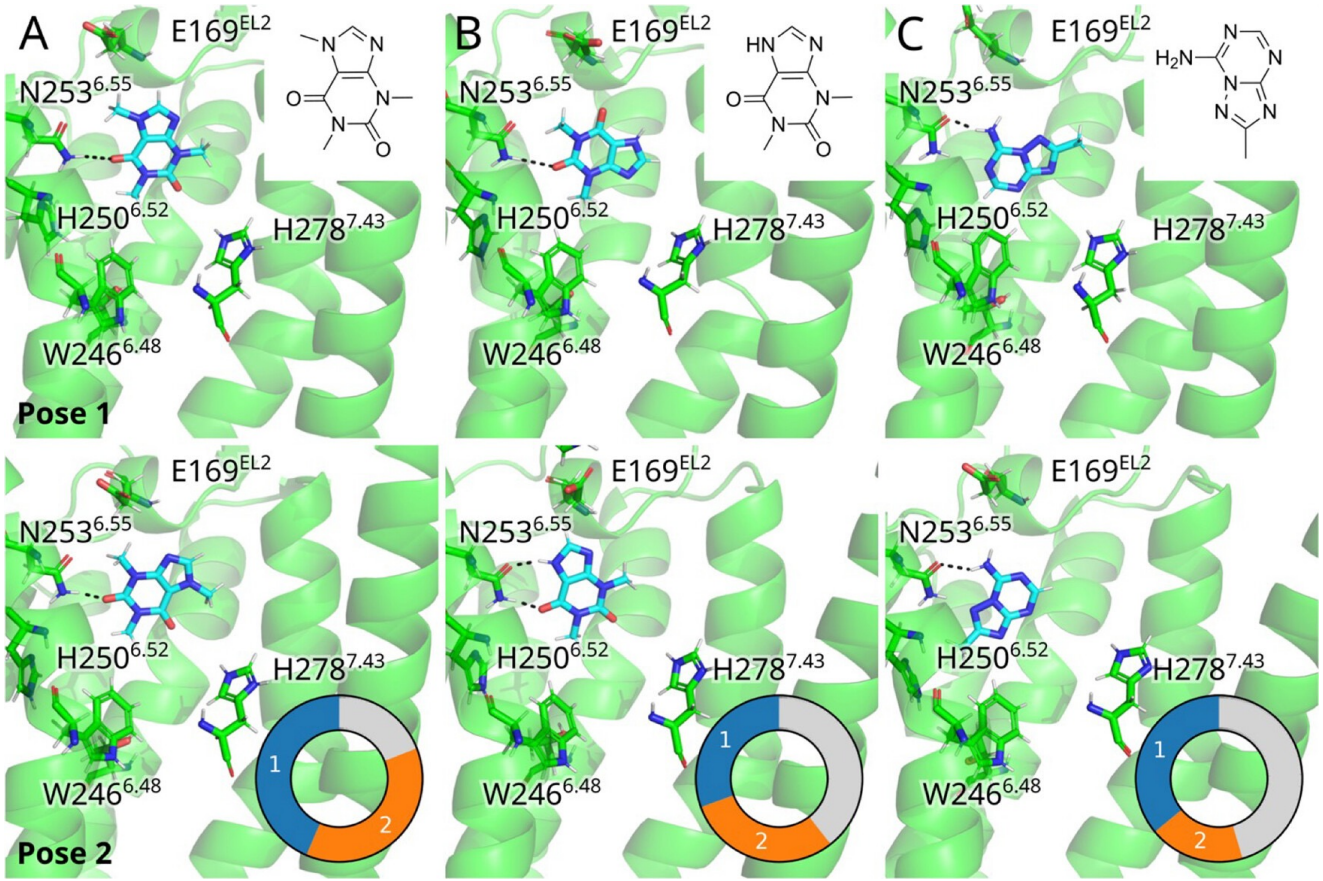


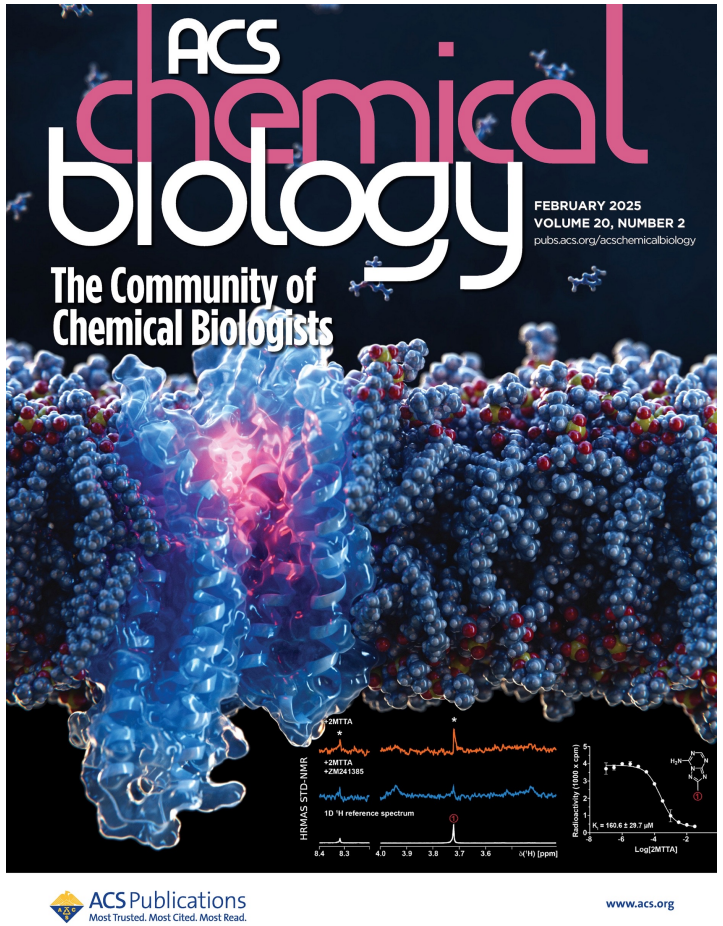






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# DOI: 10.1021/acschembio.4c00686

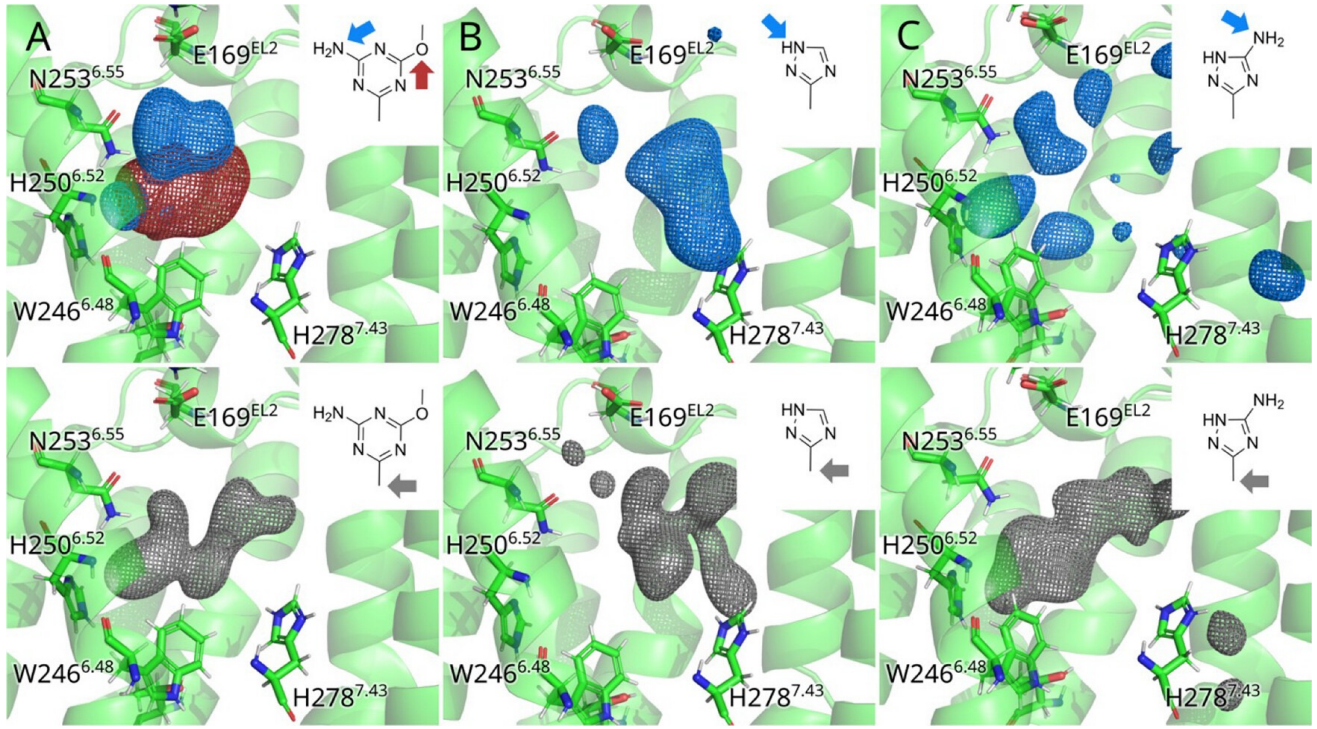





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# DOI: 10.1021/acschembio.4c00686









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# Thank you for your attention!